

“Vigilant Eye”: An Intelligent Pattern-Recognition Framework for Weapon Detection in CCTV Surveillance Systems

The Major Project report submitted for the partial fulfilment of the requirements of the Degree of

**Bachelor of Technology (B. Tech) in
Computer Science and Engineering**

By

SNEHAMOY GHOSH

Reg No – 002-103-2022-173

Roll No – 002-BCS-2022-005

Under the Supervision of

Dr. Payal Bose



Department of Computer Science and Engineering

SWAMI VIVEKANANDA UNIVERSITY

A Project Report On

**“VIGILANT EYE”: AN INTELLIGENT
PATTERN-RECOGNITION FRAMEWORK FOR
WEAPON DETECTION IN CCTV
SURVEILLANCE SYSTEMS**

Submitted By

Name	Roll No
ANANYA ROY	002-BCS-2022-022
INDRANIL PAUL	002-BCS-2022-174
ROHITASHWA PAL DAS	002-BCS-2022-203
SANKHADEEP BHOWMIK	002-BCS-2022-020
SNEHAMOY GHOSH	002-BCS-2022-005

Department- CSE, Semester- 8th

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Under the guidance of

Dr. Payal Bose

A Project Report

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Department of Computer Science and Engineering

SWAMI VIVEKANANDA UNIVERSITY

CERTIFICATE

This is to certify that the project entitled: **“VIGILANT EYE”: AN INTELLIGENT PATTERN-RECOGNITION FRAMEWORK FOR WEAPON DETECTION IN CCTV SURVEILLANCE SYSTEMS** submitted to SWAMI VIVEKANANDA UNIVERISTY in the partial fulfilment of the requirement for the award of the B. TECH degree in COMPUTER SCIENCE AND ENGINEERING of Project-II (BTCSS881) is carried out by

ANANYA ROY ***002-BCS-2022-022***

INDRANIL PAUL ***002-BCS-2022-174***

ROHITASHWA PAL DAS ***002-BCS-2022-203***

SANKHADEEP BHOWMIK ***002-BCS-2022-020***

SNEHAMOY GHOSH ***002-BCS-2022-005***

Under my guidance. The matter embodied in this project is genuine work done by the students and has not been submitted whether to this University or to any other University/Institute for the fulfilment of the requirement of any course of study.

(Project Supervisor)

Dr. Payal Bose

Swami Vivekananda University

Barrackpore, WB

STATEMENT BY THE CANDIDATES

ANANYA ROY *002-BCS-2022-022*

INDRANIL PAUL *002-BCS-2022-174*

ROHITASHWA PAL DAS *002-BCS-2022-203*

SANKHADEEP BHOWMIK *002-BCS-2022-020*

SNEHAMOY GHOSH *002-BCS-2022-005*

B. Tech 8th Semester

Dept. of **Computer Science and Engineering**

Swami Vivekananda University

We hereby state that the Project Report entitled “**VIGILANT EYE**”: **AN INTELLIGENT PATTERN-RECOGNITION FRAMEWORK FOR WEAPON DETECTION IN CCTV SURVEILLANCE SYSTEMS** has been prepared by us to fulfil the requirements of **Project-II (BTCSS881)** during the period January 2024 to June 2026.

Signature of the students

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ABSTRACT

The recent surge in urban sprawl and the escalating security issues have exposed weaknesses associated with the conventional CCTV surveillance systems, which are mainly based on manual surveillance and reactive surveillance. The study suggests a project titled Vigilant Eye, which is a smart pattern-recognition system that will be used to improve CCTV surveillance by applying both artificial intelligence and computer vision algorithms. The main aim of the research is to create a real-time weapon detection system that can enhance situational awareness and allow it to detect the threat in advance. The offered framework is a combination of object detection named as Deep Learning (DL)-based on YOLOv8 and object tracking based on the Deep SORT algorithm. Besides, they are supplemented with behavioural analysis and anomaly detection mechanisms to enhance the contextual understanding and minimize false alarms. A combination of publicly available datasets and custom CCTV-like data is used to train and test the system and guarantee its strength in the real world. Experimental findings indicate the system has high detection rates of about 90% and also has low latency (Less than 100 milliseconds per frame), which makes it applicable in real-time surveillance. The contextual analysis and tracking greatly boost the detection reliability relative to individual models. Also, the paper relates to ethical aspects, including privacy protection by using anonymization and edge processing tools. In general, these results suggest that AI-enabled surveillance systems can become the means of greatly enhancing the efficiency, accuracy, and responsiveness of security infrastructures in the present-day era. The Vigilant Eye framework proposed here is a cost-efficient and scalable method of intelligent surveillance in a smart city, transport station, or public places.

Keywords: *Artificial Intelligence (AI), CCTV Surveillance, Weapon Detection, YOLOv8, Deep SORT, Computer Vision, Real-Time Monitoring, Intelligent Surveillance Systems, Smart Security.*

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CHAPTER 1

INTRODUCTION

Chapter 1

INTRODUCTION

1.1 Background of CCTV Surveillance

Close Circuit Television (CCTV) surveillance systems have increasingly become a critical part of the contemporary security systems in cities, transport networks, institutions of learning, business centres, and homes. CCTV systems have traditionally been created to do passive monitoring, where the video footage is logged continuously and the footage is viewed manually after an event has taken place. This is a reactive mode of operation that restricts the efficiency of surveillance systems in thwarting crimes since possible threats are usually realized once damage or loss has occurred. With the growth of cities and population density, the amount of surveillance data collected by thousands of cameras has surpassed the capacity of human operators to view the feeds effectively in real time.

The fast-growing popularity of Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies has introduced new possibilities to convert the traditional CCTV networks into smart surveillance systems that can be analysed automatically [23]. These technologies allow systems to identify objects, patterns, monitor people and identify abnormal behaviour without the need to have human supervision. Intelligent surveillance is not only meant to record but also to interpret the occurrence of an activity and therefore, proactive intervention can be made. This change of passive observation to a situation-awareness is especially valuable in places of heavy people such as railway stations, airports, marketplaces and campuses where manual observation is likely to be affected by human fatigue, slow reaction and oversight.

Moreover, intelligent surveillance becomes cost-effective to be implemented on large-scale CCTV architecture because the incorporation of AI with the current CCTV system makes it unnecessary to upgrade hardware at high costs [24]. The introduction of edge computing and privacy-saving systems has also taken into account the issue of data

latency and unauthorized access. As a result, according to the recent study by Haritha *et al.* (2025)[25], AI-based CCTV systems can be regarded as one of the crucial changes in security management as it allows real-time threat detection, enhances the level of public safety, and allows using resources more efficiently. It is the basis of the development of smart and automated crime prevention systems based on technological change.

1.2 Problem Statement

Despite the fact that CCTV cameras are extensively deployed in the city and homes, the majority of the surveillance systems that are currently present are reactive in nature and require constant human supervision. The security operators need to monitor several video streams at a time, which can easily cause fatigue, lack of attention and the inability to identify serious threats like weapons or violent behaviour [26]. Thus, dangerous incidents are often detected once they happen, which limits the possibility of the surveillance system to prevent crime or react promptly to an emergency.

The huge amount of video data produced on a daily basis is another big problem. Massive footage has not been checked to avoid the occurrence of an incident; therefore, it is sluggish and inefficient [27]. The conventional rule-based systems are also not able to adjust to the reality on the ground like low lighting conditions, congested areas and dynamic human behaviour and tend to generate a large number of false alarms. Thus, the objective of “Vigilant Eye” is to overcome these problems by creating a smarter, more precise, and privacy-conscious weapon detection system based on the existing CCTV systems.

1.3 Research Aim and Objectives

1.3.1 Research Aim

The main aim of the specified investigation is to create an AI-powered surveillance system that is an addition to the existing CCTV system by providing the appropriate level of weapon recognition and automatic interpretation of the threat. The study is geared towards transformation of the traditional surveillance which is largely reactive and based on manual monitoring to an active and efficient security entrance. The suggested system will accomplish the role of identifying suspicious activities with the

least number of false alarms and greater accuracy through integrating the functions of computer vision, such as object detection, multi-object tracking, pose estimation, and anomaly detection.

Furthermore, the research will ensure that the framework will be compatible with the existing CCTV systems without any hardware upgrading to get the framework to work. Overall, the paper is supposed to offer a viable, scalable and privacy-friendly intelligent surveillance system, which can be applied in the real world.

1.3.2 Research Objectives

To achieve this aim, the following research objectives are selected:

RO1: To critically evaluate the drawbacks of traditional CCTV surveillance devices, in particular, presence of manual monitoring systems and delay in investigations.

RO2: To design and implement an object detection, multi-object tracking, pose estimation and behaviour analysis modular AI-based system to support real-time weapon detection.

RO3: This is to integrate machine learning-based anomaly detection with rule-based logic to reduce the occurrence of false alarms and enhance the contextual accuracy.

RO4: The performance of the system can be compared regarding the measures such as precision, recall, mAP, tracking accuracy and latency to conclude that it is possible in real time.

RO5: In order to be able to have responsible and ethical deployment, privacy-concerned mechanisms, including edge processing and selective anonymization, need to be deployed.

1.4 Research Hypothesis

In order to guarantee the conceptual and methodological conformity, every hypothesis is directly correlated with the set research objectives:

H1: The AI-based surveillance systems will exhibit statistically significant increase in reaction time and accuracy in detection more than the conventional human operated CCTV surveillance systems.

H2: Object detection, multi-object tracking, pose estimation and behaviour analysis will be of great help in improving the accuracy of the contextual weapon detection compared to when each strategy is used separately.

H3: Hybrid implementation of machine learning-based anomaly detection strategy and rule-based contextual modelling will greatly cut down false positives and retain high sensitivity in detection.

H4: The suggested modular AI system will have the real-time operation, and inference latency will be kept below the predetermined limits (less than 100 ms per frame) without impairing accuracy or recall.

H5: Edge processing plus selective anonymization will greatly decrease the amount of bandwidth used and amount of risk in privacy exposure low enough that the detection performance does not decline significantly.

H0 (Null): There will be no statistically significant differences between the proposed system and the traditional CCTV systems in the performance measures considered.

1.5 Importance of the Study

The research is relevant in that it will discuss an increasing demand for smart and active security measures in more intricate urban settings. With the increasingly dynamic crime patterns and the increasing use of the public spaces, the traditional surveillance mechanisms that solely use manual surveillance tools have become inadequate in providing adequate time in detecting and reacting accordingly. This study will help in creating more intelligent surveillance systems that can detect threats in real-time and aid in preventing measures, instead of analyzing them after the event through integrating AI with the existing CCTV infrastructure [34].

Technologically, the research shows that object detection, tracking, and pose estimation coupled with anomaly detection can enhance situational awareness and minimize the possibility of missed events [35]. The framework also suggests the significance of

minimizing false positives by the use of contextual analysis, which is pivotal in keeping operator confidence and making successful decisions. In addition, the focus on modular architecture and the ability to integrate with the existing network of cameras will render the solution affordable and scalable to large-scale applications in a public institution, transportation networks, and smart cities.

The study also has practical implications to the law enforcing agencies, security personnel and policy makers through offering a systematic way of carrying out intelligent surveillance and ensuring all operations remain effective. Notably, the research takes into account such ethical issues as privacy protection, data minimization, and controlled access, which is very relevant to the concerns about the abuse of surveillance technologies [36]. The research will help balance between the personal rights and the safety of people by encouraging responsible use of AI.

In general, the proposed study contributes to the development of the sphere of intelligent video analytics, as it suggests a well-developed, efficient, and ethically justified surveillance system that will allow preventing crimes and responding to emergency situations more effectively.

1.6 Structure of the Research Paper

The proposed Research Paper has been logically divided into seven chapters to give a smooth flow of ideas starting with the conceptual base to technical proposal and analytical analysis of the proposed paradigm of intelligent CCTV surveillance system, “Vigilant Eye”.

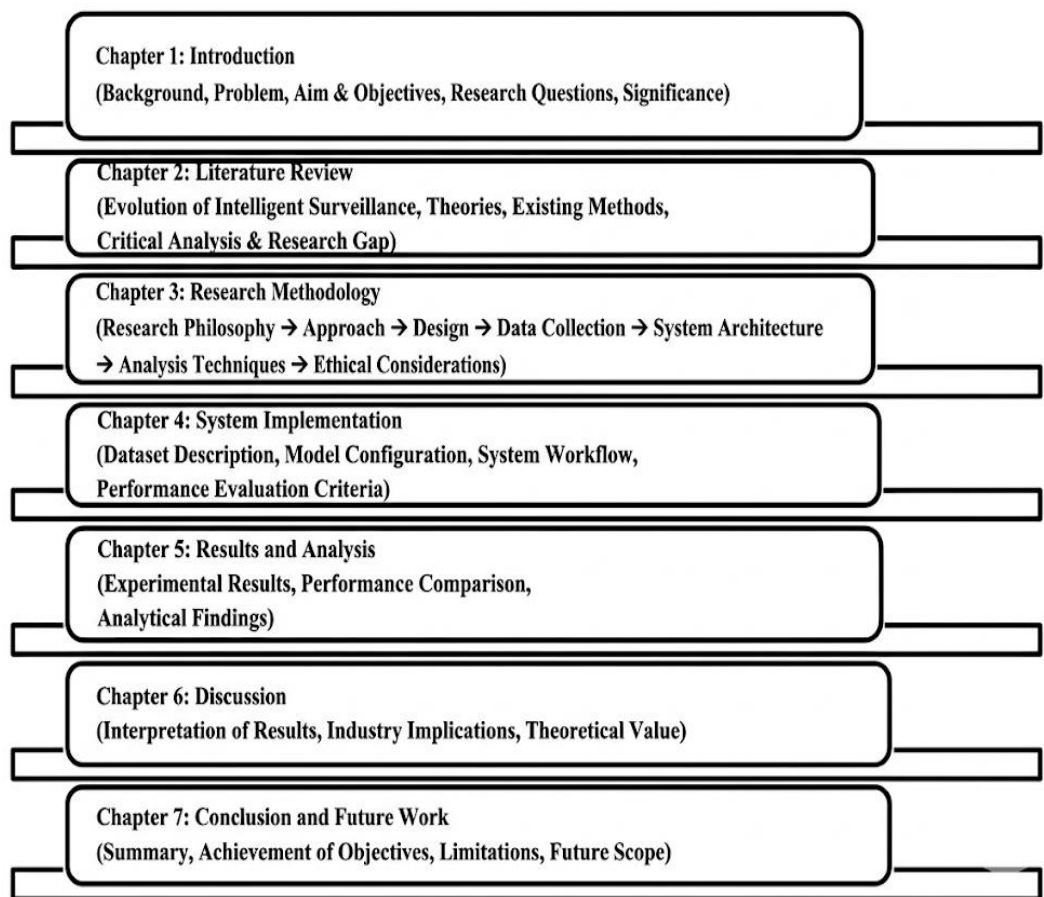


Figure 1.1: Structure of the Research Paper

The research is started by the Abstract containing a brief description of the research purpose and methodology and the main contribution of the research. Chapter One presents the background of CCTV surveillance, the main issue, the purpose and objectives of the research and the importance of incorporating artificial intelligence into the currently existing monitoring systems.

Chapter Two will provide a literature review of the development of intelligent surveillance, the existing theoretical frameworks, and recent technical approaches. It critically discusses the current methods and shows the research gap that would warrant the proposed system.

Chapter Three presents the research methodology, such as the philosophical position, research strategy, design plan, data collection process, system architecture and analytical tools and ethical issues that will be involved in the formulation of the framework.

The fourth chapter is devoted to the implementation of the systems, where the datasets, the model configuration, the workflow of the operations and the evaluation parameters are described. Chapter Five gives the research findings and performance analysis, and Chapter Six gives an interpretation of findings and a discussion of their practical and theoretical consequences to the surveillance field.

Lastly, Chapter Seven is the final chapter of the dissertation, which summarizes the contributions, evaluates the objective achievement, states limitations and formulates the future research directions.

CHAPTER 2

LITERATURE OVERVIEW

Chapter 2

LITERATURE OVERVIEW

The recent development of Artificial Intelligence (AI) and Machine Learning (ML) has greatly changed the surveillance system. The emphasis has been placed on both active video capture and passive video capture and smart real time analysis of the video network to detect anomalies, crime prediction and proactive response in CCTV networks [23]. We have seen that multiple existing studies have shown progress in weapon detection and video anomaly analysis, but they still struggle in many aspects of real world problems. Many struggles with low-resolution footage, crowded scenes, or insufficient lighting [1], while others cannot detect small or partially hidden weapons [20]. Many models are computationally heavy for real-time use but they lack accuracy that leads to false alarms [28]. These drawbacks show that there is a need for a more accurate, real-time system that will work on existing CCTV infrastructure. We are aiming to develop this project specifically to overcome these limitations. Our project will help to overcome these drawbacks by detecting even minor weapons, reducing false alarms, and respecting the privacy of people [32].

2.1 Evolution and Theoretical Framework

The development of CCTV surveillance technology is indicative of a larger change in the manner in which communities are conceptualizing security, risk and technological competency. Since their dawn as passive recording media all the way to the current AI-enhanced analytic tools, CCTV networks have experienced a paradigm shift, which reflects the shift of the manual observation to the algorithmic intelligence and, thus, the need to have a thorough understanding of this paradigm change to place the so-called Vigilant Eye framework proposed into the context of its scholarly and historical context [27].

2.1.1 Early Surveillance:

CCTV originated as crude sets of video recording devices that were invented to record their visual evidence. In their initial realization, they took the form of closed-loop systems in which the cameras relayed an image to monitors which were controlled by humans. Such settings were naturally responsive; the main task of such settings was not prevention but reporting. The evaluation of footage was often conducted after an incident had happened, and it served in the capacity of a preventative device instead of a prohibitory one [1].

Early surveillance systems were based on a theoretical basis that was in the theory of deterrence. It was believed that the presence of cameras would deter crime but this belief was based on visibility and publicity and not an analytical capability in real-time. Its operators could not watch more than three video channels at a time, a task that rendered them fatigued, cognitive overloaded, and supervisory. Empirical studies in security operations are constantly reproducing the same findings that manual observation reduces the level of attention over time hence exposing one to more chances of failing to spot a threat [21]. With the proliferation of cities and the consequent growth in the quantity of cameras being installed, the drawbacks of human-centric surveillance became more pronounced. The rapid implementation of the cameras in large metropolitan areas enables thousands of cameras to produce huge amounts of video data every day. The amount of data exceeded the ability of human operators to process it, leading to what is commonly known as the surveillance paradox: the more cameras one has, the more they do not always lead to a security improvement when the data cannot be processed [27].

2.1.2 Digitizing and Networked Surveillance:

The next significant step in the history of CCTV systems was the shift to a digital recording. DVRs and later Network Video Recorders (NVRs) enabled an increase in storage capacity, remote accessibility, and the interrelation with IP based systems. This change made it more scalable and more accessible but without a novel change in the fundamentals of reactive surveillance. This was the point when rule-based video analytics began to develop. It was pioneered with motion-detection algorithms, tripwire notifications and predefined zone coverage to perform automated surveillance on

specific aspects [14]. These systems were based on deterministic programming: an alert was given when a pixel changed above a value. Although novel at the time, they were rule-based systems that never knew the context, they used to detect motion but not sense. As an illustration, they were unable to make a difference between an inoffensive object and a weapon, or make sense out of behavioural clues.

This stage is a kind of an initial version of computer-aided help but not intelligence. Theoretical debates of the day started to consider the idea of computational surveillance, that is, machines might be used to supplement human perception but since machine-learning techniques were not as robust as they were today, practice could not have been as expansive [52].

2.1.3 Birth of Machine Learning and Computer Vision:

The actual revolution became realised along with the development of machine learning (ML) and deep learning. The use of convolutional neural networks (CNNs) is a breakthrough in computer vision since it is a contrast to the rule-based models which are learned by hierarchical features directly through the data [3]. They do not have to be taught patterns, but as they interact with large datasets they generate internal representations. This paradigm was marked by a changed theoretical position: data-driven intelligence. Surveillance systems were now able to go beyond the detection of movements and towards identification of objects, characterization of behaviours and forecasting an anomaly. The frameworks of object-detection (R- CNN, Fast R- CNN, Faster R- CNN, SSD and finally, YOLO (You Only Look Once) made it possible to detect objects in video streams in real time. YOLO-based networks, especially in subsequent versions like YOLOv8, represent the next generation of the former, moving off a multi-stage detection pipeline to a single-stage, unified, detection model with real-time inference abilities [57]. These models manipulate the entire image at a single pass, which significantly lowers the latency, but at high accuracy. This advancement is essential where immediate response is required in the sphere of surveillance.

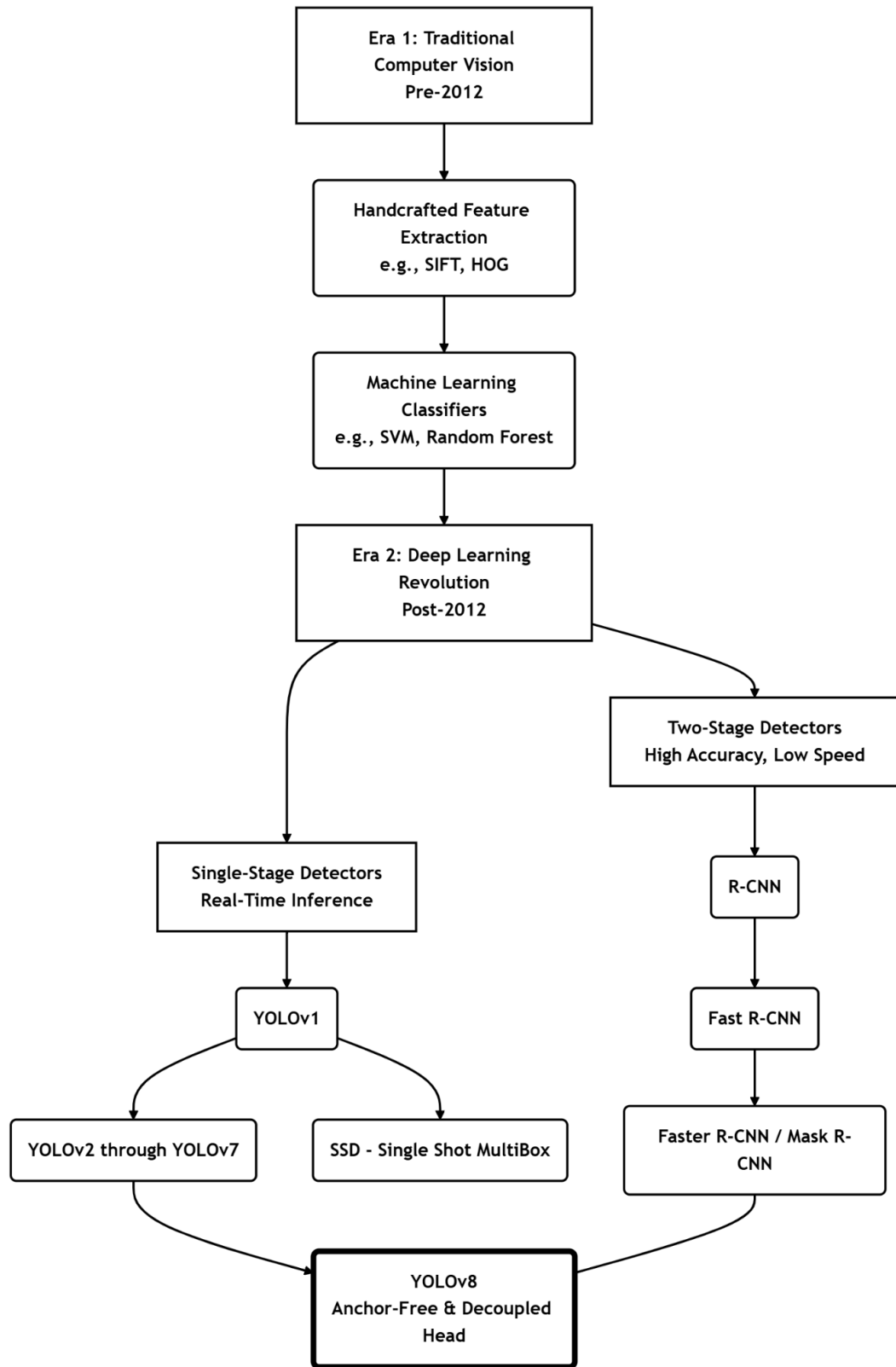


Figure 2.1: The Evolution of Object Detection Models

The conceptual model of the modern intelligent surveillance systems integrates several spheres:

- Computer Vision Theory- allowing machines to obtain important visual information.
- Statistical Learning Theory - helps in generalisation based on training data to the real world.
- Anomaly Detection Theory - determining deviation of learned regular patterns.
- Human-Machine Interaction Theory - maximising the co-operation of AI systems and operators.

This inter-disciplinary intersection has transformed surveillance into a proactive and predictive and not a reactive system [35].

2.1.4 Context Based Intelligence:

Although the development of object detection was a significant progress, modern studies focus on situational consciousness. Weapon detection is not just about identifying the shape of the weapon, but is concerned with the analysis of the environment, the posture, the movement pattern, and even the situational analysis [52]. Multi-object tracking algorithms, behaviour-analysis systems and pose-estimation models help in understanding on a higher level. As an example, it is quite different to detect a gun on a holster of a security officer and it is quite different to detect a gun raised in an aggressive posture in a large population. Contextualization minimizes false positives and maximizes accuracy of decision making. The paradigm shift in this case is the move between the pixel recognition and semantic deciphering. Smart surveillance now represents spatial-temporal modelling in which space (location in the frame) and time (flow across frames) have a causative influence in interpretation [48]. The temporal convolutional networks and attention mechanisms allow systems to understand the motion dynamics, as opposed to a single frame. This development is a sample of a larger AI development that moves ahead to the processing of cognition. Even though this is not comparable to the reasoning in humans, current AI systems simulate contextual judgement through the integration of various signals.

2.1.5 Predictive Security and Algorithms Intelligence:

Predictive intelligence is the recent development in surveillance. In addition to identifying the current threats, systems are becoming more and more attempts to predict risk. Pattern -recognition systems, criminal forecasting and crime prediction models use past data to predict possible hotspots or behavioural antecedents.

Some of the criminological theories used in predictive surveillance are routine-activity theory and environmental criminology. Opting to apply historical crime data, spatial analysis, and real-time video feeds, the algorithmic systems attempt to predict the places where events can happen.

However, predictive models generate ethical and methodological issues. Training data has biases which can cause unequal focus on particular locations or populations. In turn, the principles of responsible AI have been reflected in theoretical frameworks and focus on fairness, transparency, and accountability [17].

2.1.6 Edge Computing and Privacy Architecture:

Another important theoretical innovation is adopting edge computing paradigms. Earlier surveillance systems were based on the principle of centralized server based, effectively making the systems susceptible to high latency and privacy invasion. The edge-based artificial intelligence mechanisms process the visual data at the device or very closely to the source of data, thus reducing the need to transmit data and preventing the possibility of exposure [7].

This radical change is in line with theories that seek to conserve privacy in machine learning. Anonymization methods, data selective retention, and on-board inference machines are some of the methods used to minimize exposure of sensitive information in the process of surveillance [66].

The theoretical prism used in this paper goes beyond the standard performance indicators and also incorporates the issue of ethics and state apparatus. Based on this, the assessment of intelligent surveillance systems is not only done based on the indices of accuracy but also conformity to the standard of data protection as well as the general behaviours of people in society.

2.1.7 Collaboration between Human and AI in Surveillance:

Even though technological advancements have been significant, human control plays a crucial part. The most effective systems will be in hybrid mode where artificial intelligence will carry out constant surveillance and report irregularities, and human operators will verify the irregularities. Such phenomena as alert fatigue are addressed in this collaborative paradigm. The use of artificial intelligence in reducing cognitive load through the irrelevant hunt of footage is supported by human operators as the operators provide context and ethical consideration. The theory of augmented intelligence, as opposed to total automation, is the basis of the rationale behind such an approach.

2.2. Case Studies and Technical Methodologies

Over the last few years, there has been an increase in the practical application of artificial intelligence in a closed-circuit television surveillance environment [29]. In a variety of case studies, the combination of deep learning measures, anomaly detection algorithms, and real-time inference models in security architectures is demonstrated and, as such, can provide solutions to both technical and operational issues.

2.2.1 Weapon Detection Systems using YOLO:

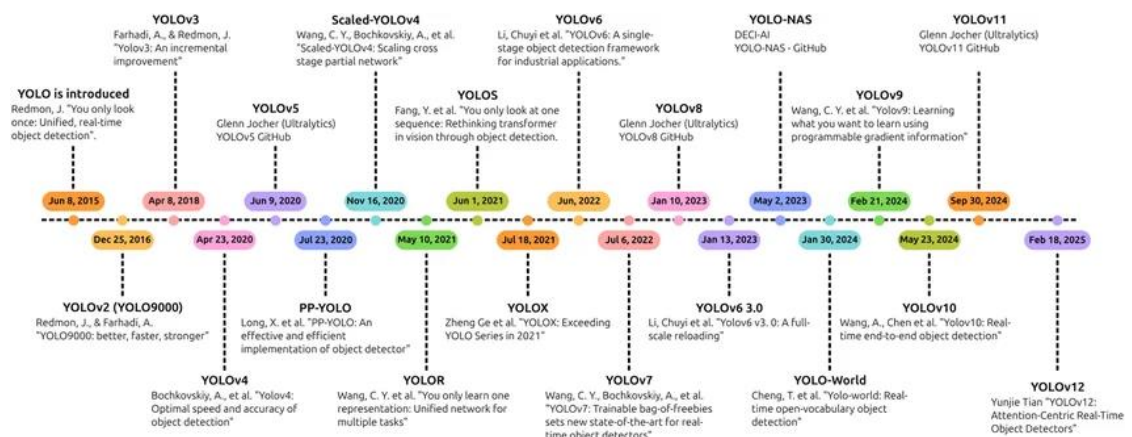


Figure 2.1: Evaluation YOLO over time

(Source: Raza *et al.*, 2025)

Architectures made using YOLO have taken the stage in surface surveillance studies. According to empirical studies, Wang *et al.* (2023) [39] confirmed that real-time

detection based on contemporary YOLO architecture can perform at the state of the art and with high accuracy using optimum inference speed under different environmental factors in diverse environmental settings. The single-stage detectors pipeline minimizes the computational load of the multi-stage detectors. Technically the input image is divided in grids and block-wise each grid is predicted to have a bounding box and a probabilistic score of the image. Its anchor-free detection head coupled with an enhanced network of feature pyramids enhances detecting small objects- a feature which is largely relevant towards detecting hidden or partly recorded weapons.

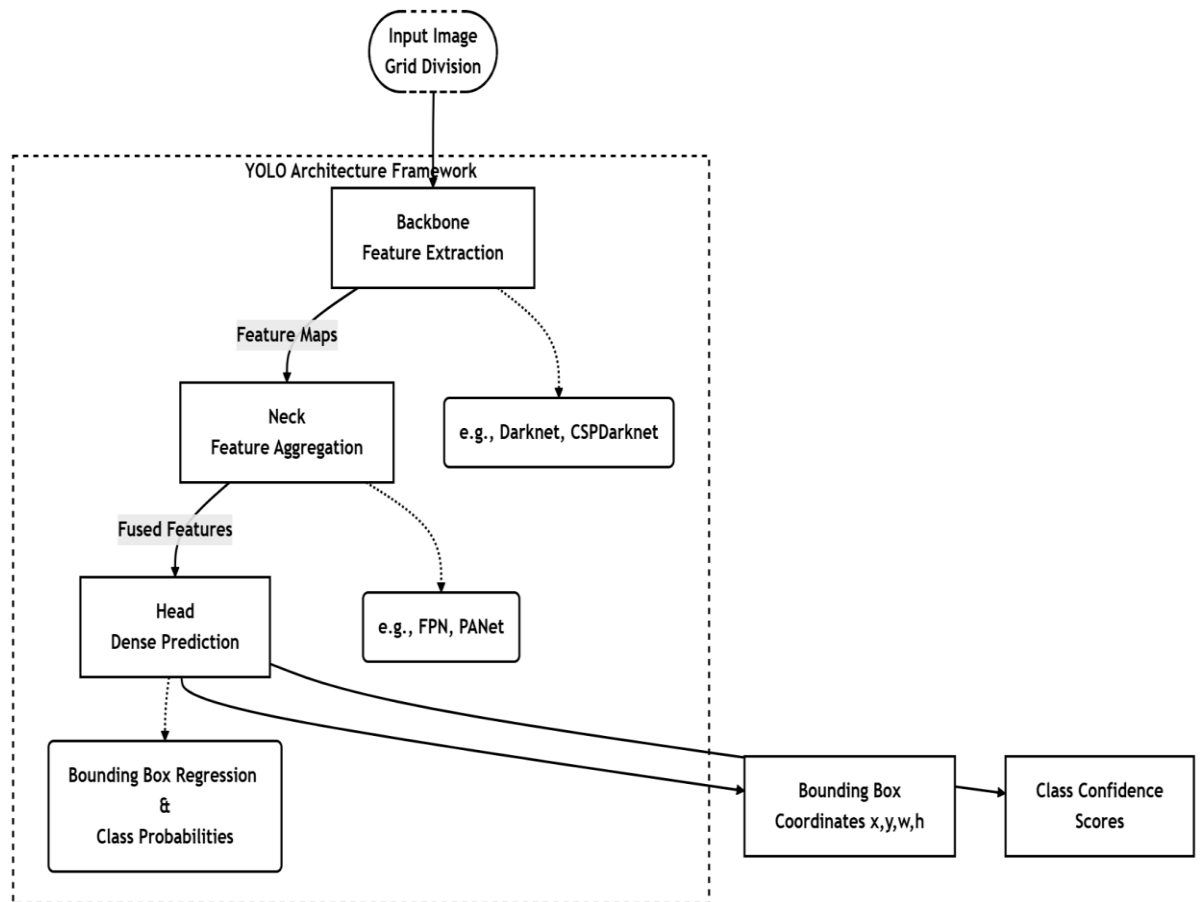


Figure 2.2: Fundamental Architecture of YOLO Models

Transfer learning is also used on a regular basis to tune the retrained YOLO models on dataset-specific models. It is effective in terms of less time of training and higher generalization performance, particularly the use of labelled weapon datasets that are not large.

2.2.2 Multi-Stream and CNN based Hybrid Models:

Wu *et al.* (2021) [47] came up with a framework of weakly-supervised spatio-temporal anomaly detection that combines spatial feature extraction and temporal reasoning to boost the detection strength in surveillance scenes. Hybrid systems combine the convolutional neural network feature extraction abilities together with regression based localization systems. Two-stream neural networks Two-stream neural networks (such as inflated 3D I3D convolutional neural networks) combine two streams of processing: spatial and temporal. Frame-level features are processed in the spatial stream and optical flow is processed in the temporal stream to obtain the motion dynamics. These two-stream models show effectiveness in the detection of violent behaviour as opposed to fixed objects only. However, multi stream architectures add to complexity. Live deployment often requires either edge-stitching or concerted edge-optimisation plans with the assistance provided by a computer graphics processor(s) [28].

2.2.3 Crowd Scene Model and Anomaly Detection Model:

A comprehensive review of the literature in the area of anomaly detection in crowds is presented by Ramachandra *et al.* (2020) [48], who identify such problems as occlusion, background clutter, and domain generalization. Close-up shots pose very different backgrounds, such as occlusion and overlapping of objects. In these scenarios, detection sensitivity is made sensitive to the application of attention mechanisms and through the use of the feature-fusion layers. Detection models of anomalies often make use of unsupervised or few-supervised learning paradigms. Systems do not have to undergo tedious labelling of all the aberrant events, but they do acquire normative patterns and consequently identify violations. This unmonitored method is beneficial to decline or unexpected events. Nevertheless, weak supervision can create instability where anomalies have a significant difference in form. The strength of the models used depends on the diversity of the training set.

2.2.4 Implementations of Edge Computing:

To overcome the issue of scalability and latency, Patrikar *et al.* (2022) [49], studied edge-based anomaly detection. Running models on edge units like Raspberry Pi or any other device with a graphical processing unit minimizes the need to have central servers. Edge deployment requires model compression methods, such as quantisation and

pruning, to reduce the size of the model and computational requirements whilst doing the best to preserve its accuracy. Edge/based artificial intelligence also provides greater privacy because it limits the scale of data transfers between devices; it is detection metadata that is transmitted, instead of complete video streams, to centralised systems [7].

2.2.5 Mechanisms of attention and quality of images:

Cui *et al.* (2024) [50], suggested combining image -type evaluation with attention -control systems. Poor and low-resolution images compromise detection accuracy. Pre-treatment tasks like contrast optimization, demonising, and super-resolution quality in the input image are used to improve the quality of input images before inference. Attention layers help the network to focus on salient parts in a frame and therefore enhance the detection accuracy in the cluttered backgrounds.

2.2.6 Multi-Camera Re-identification and Multi-Camera Integration.

High-rise surveillance systems combine numerous camera outputs to follow a person in spatially dispersed places. Re-identification algorithms compare feature descriptors between frames and cameras and can be used to improve situational awareness, but at the same time, other concerns of privacy and computation arise [44].

2.2.7 Crime Prediction Systems:

In addition to object detection, predictive analytics models use spatial mapping forecasts of crime historical data together with historiography. Machine-learning provides high-risk areas, and time-resolved criminal behaviour [35]. However, predictive systems are still controversial as they have the contribution of bias and ethical considerations [17].

2.3 Research Gap

Despite the recent significant progress in AI-based surveillance systems, most of the research currently tends to aim at enhancing technological standards (detection rates, mean average precision, tracking statistics, etc.) instead of the more practical question of the improvement of currently existing CCTV networks in a natural setting [57]. Most of these solutions require the implementation of multi-purpose cameras, high-density sensors or hardware intensive in nature making large-scale implementation expensive,

impracticable and not feasible to apply to public authorities or even smaller organizations with limited budgets [7]. Moreover, a few surveillance systems only use rule-based monitoring systems or machine learning models, but do not combine the two with each other [14]. Such failure to integrate can lead to high false alarms, low alert interpretation levels and lack of trust to the system by operators [26]. Another significant issue is privacy, ethical, and legal, because most systems are not designed to properly manage the risk of abuse, biased recognition, and consent violation and not many of them offer elaborate anonymization, on-machine processing, or safes storage mechanisms to protect against this risk [32]. In addition, real-world application exhibits special conditions, low-light scenarios, occlusion, overcrowding or dynamic scenes, and unpredictable human behaviour, which traditional models are not very friendly in generic terms, making their application only challenged in practice [20]. The proposed Vigilant Eye system should address these drawbacks by utilizing the existing CCTV system and reducing the necessity to introduce new hardware, integrating reusable AI functionally (object detection, multi-object tracking, pose estimation, and anomaly scoring) [57]. By combining both a rule-based logic with machine learning-based anomaly detection, the system can not only increase the accuracy and timeliness of alerts, but also improve the interpretability and context-sensitivity and, thus, reduce false positives and deliver more actionable insights [42]. Moreover, Vigilant Eye will add privacy enhancing features, such as on-edge processing, selective anonymization, and controlled access to sensitive information, which, altogether, will enable the system to act responsibly, as well as to be real-time responsive and operate across an assortment of environments and situations [65].

2.3.1 Potential Research Gaps:

- Functioning in a crowded space: Existing AI-driven surveillance technologies tend to perform poorly in highly crowded spaces, like trains stations, stadiums or festivals, where detecting, tracking and identification is challenging [48].
- False Alarms: In spite of the improvement most systems produce false alarms when exposed to unusual non-hazardous situations, insufficient lighting or background noises [28].

- Privacy concerns: Although simple edge processing has been executed, there is little usage of algorithms such as differential privacy and homomorphic encryption. The integration of these strategies might guarantee the enhanced data protection and adherence to the legal framework and ethics without decreasing system performance [32].
- Scalability on Multi-Camera Networks: The available systems are tested on single camera or small networks. To realize the real time processing, bandwidth control and the consistency in the detection of anomalies, scaling AI-driven surveillance to city-wide CCTV systems and various cameras and mixed environments is a challenge [10].

Table 2.1: Comparison of existing approaches

Author and Year	Methodology	Key Results	Limitations
Wang <i>et al.</i> (2023) [39]	YOLO V7 based on Real time object detection	The present study is state of art in terms of accuracy in real-time detection by using a sophisticated training approach, and an effective architecture design.	Tests of surveillance-type data are established to be high- Records. Nevertheless, the performance of the method suffers in such settings of extreme occlusion and very dense crowd situations.

Terven & Córdova-Esparza (2023) [40]	Strong review of YOLO Architectures	The systematic summary of YOLO variants (v1-v8) is reported, where the attention is devoted to the architectural improvements that have been performed on the detection speed and precision of the architecture in order to be useful in the usage of the accelerated versions of the system in real-time.	Although the review provides theoretical knowledge, it lacks empirical validation that is conducted by deployment of experiments.
Doshi & Yilmaz (2020) [41]	Constant Learning of Anomaly Detection.	An improvement of the adaptability is suggested through a continuous learning structure model to be used in anomaly identification in surveillance scenarios.	The strategy requires careful tuning in order to reduce the possibility of disastrous forgetfulness.
Chalapathy & Chawla (2021) [42]	Deep Learning for Anomaly Detection	A literature review provides an overview of deep learning methods of abnormal entity detection in	It emphasizes that a number of these models are modelled on the basis of huge datasets with labels

		various fields, such as video surveillance.	and are expensive to compute.
Xu <i>et al.</i> (2020) [43]	Graph-Based Video Anomaly Detection	The graph convolutional networks are used to describe the spatial-temporal correlations of the video information.	The enhanced architectural character can hamper real time feasibility.
Ge <i>et al.</i> (2020) [44]	Unsupervised Domain Adaptation for Person Re-Identification	Pseudo-Label refinement strategies are used to increase blurry camera tracking and re-identification.	Although the performance can be optimized, the approach affects a high level of computation requirements in implementing large-scale systems.
Ristea <i>et al.</i> (2022) [45]	Video anomaly detection Self-supervised.	Spatio-temporal modelling is proposed to be self-supervised to identify anomalous events in surveillance video.	The success of such a method is much dependent on the diversity of data of training.

Mittelstadt <i>et al.</i> (2021) [46]	Algorithms which are explainable and accountable...	The paper covers tenets of transparency, fairness and accountability in artificial intelligence systems that are used in high-risk areas like surveillance.	Though an explanation of a conceptual framework is provided, the details of deployment at the implementation level are not covered.
Wu <i>et al.</i> , 2025 [47]	Weakly-Supervised Spatio-Temporal Anomaly Detection.	An anomaly detection strategy based on relationship reasoning and extensive network architecture is built that is weakly-supervised.	Based on low-resolution video-level annotations, the method can face limited opportunities when tackling complicated anomalies.
Ramachandra <i>et al.</i> (2020) [48]	Single-Scene video anomaly detector Survey.	Intensive comparison of reconstruction, prediction, and hybrid methods of detecting anomalies is given to surveillance scenarios. Trends on benchmarking, limitations in data are detected.	Lack of domain generalization and real-world scalability in most anomaly detection models is highlighted in the survey.

CHAPTER 3

DATASET DETAILS

Chapter 3

DATASET DETAILS

The proposed Vigilant Eye framework was trained and tested on a set of three publicly available datasets of weapons recorded from the Kaggle and Roboflow portals. To achieve both an improved diversity and robustness of the proposed deep learning model as well as a good generalisation ability under realistic CCTV surveillance conditions, multiple datasets were integrated together. Selected datasets include a variety of environmental conditions, camera orientations, lighting conditions, and occlusions that are crucial for enhancing the practical use of AI-powered surveillance systems in the real world. Research in recent times in the field of computer vision suggests that the use of heterogeneous datasets helps in decreasing model bias and thus enhancing the model's performance in detecting models across several different domains [4].

The selected data sets were carefully chosen to simulate real-world surveillance scenarios with weapons in a variety of settings, including cluttered backgrounds, low lighting, limited visibility and motion blur. This is important for intelligent surveillance research as visual appearances of an object in the real world are rarely clean and perfectly centred in the camera frame (Khan *et al.*, 2022). The joint data strategy also enables the transfer learning and enhanced feature generalization in various visual contexts.

3.1 Data Sources and Description

The data used for training and validation of the model were three independent datasets:

3.1.1 Dataset 1: CCTV Weapon Detection Dataset

The first set of data provided from kaggle, consists of CCTV style surveillance photos, for the purpose of identifying weapons. This dataset contains a variety of items, such as pistols and rifles, knives, and non-weapon objects, which were captured indoors and

outdoors. The images are so similar to the quality of surveillance cameras that the dataset is available for the object detection research that takes place in real-time through CCTVs. The database also includes brightness variations, object distance and camera orientations to enhance the robustness of deep learning detection models (Wang *et al.*, 2024).

The data set is available via: <https://www.kaggle.com/datasets/simuletic/cctv-weapon-dataset>

3.1.2 Dataset 2: Weapon Detection Dataset (Roboflow)

The second set is taken from Roboflow Universe and consists of annotated images of weapons to be used in YOLO object detection systems. This dataset contains a number of weapon categories and YOLO-like bounding-box annotations. A big benefit of this dataset is that it has real-world scale variations and partial occlusion of objects. These conditions enhance the performance of the model in detection of hidden or partially visible firearms in surveillance cameras' streams (Prince *et al.*, 2024).

Data can be obtained from: <https://universe.roboflow.com/yolov7test/weapon-detection-m7qso>

3.1.3 Dataset 3: Gun and Knife Detection Dataset

The third dataset was chosen, to enhance the sensitivity of the system for detecting smaller and intermediate-size handheld weapons. This dataset is mainly compiled around different interactions that can be seen between human actors and various environment conditions, with an emphasis on guns and knives. The dataset enhances the YOLOv8 framework's ability to detect the structure of metallic weapons in complex scenes with different lighting and backscatter. Some recent works have shown that the fusion of specific weapons' data leads to efficient feature extraction in deep learning-enabled surveillance systems (Shingh, 2026).

The data are available from: <https://universe.roboflow.com/mahad-ahmed/gun-and-knife-detection>

3.2 Dataset Integration Strategy

To obtain more diverse visual patterns and reduce the dependency with regard to the datasets, the three datasets were integrated into a single training repository. One of the key principles in AI-driven surveillance is to integrate datasets from different sources, as datasets from a single source are prone to overfitting and poor generalisation. The combined data set enabled the proposed model to learn the appearance of weapons in various perspectives, resolutions, size and environment.

To ensure data quality, before integration samples and corrupted images were duplicated. We converted all the annotations to YOLO format which has annotations as class labels and bounding-box coordinates in normalised form. The obtained integrated data was then split into three data sets as 80:10:10, which is for training, validation and testing of the model to ensure a balanced evaluation of the model.

3.3 Data Pre-processing and Data Augmentation

To reduce the inconsistencies of the data and achieve model stability during learning, data pre-processing was carried out. All of the images were resized to 640×640 pixels for compatibility with the YOLOv8 architecture input requirements. The image intensity distribution in the images was also normalized to standardize image intensities and promote convergence for training (Zhang *et al.*, 2025).

Several augmentation techniques were used to increase robustness to real world surveillance conditions, these were:

- Horizontal flipping
- Random rotation
- Adjusting the brightness and contrast. Brightness/Contrast adjustments.
- Mosaic augmentation
- Gaussian noise injection
- Scaling and cropping

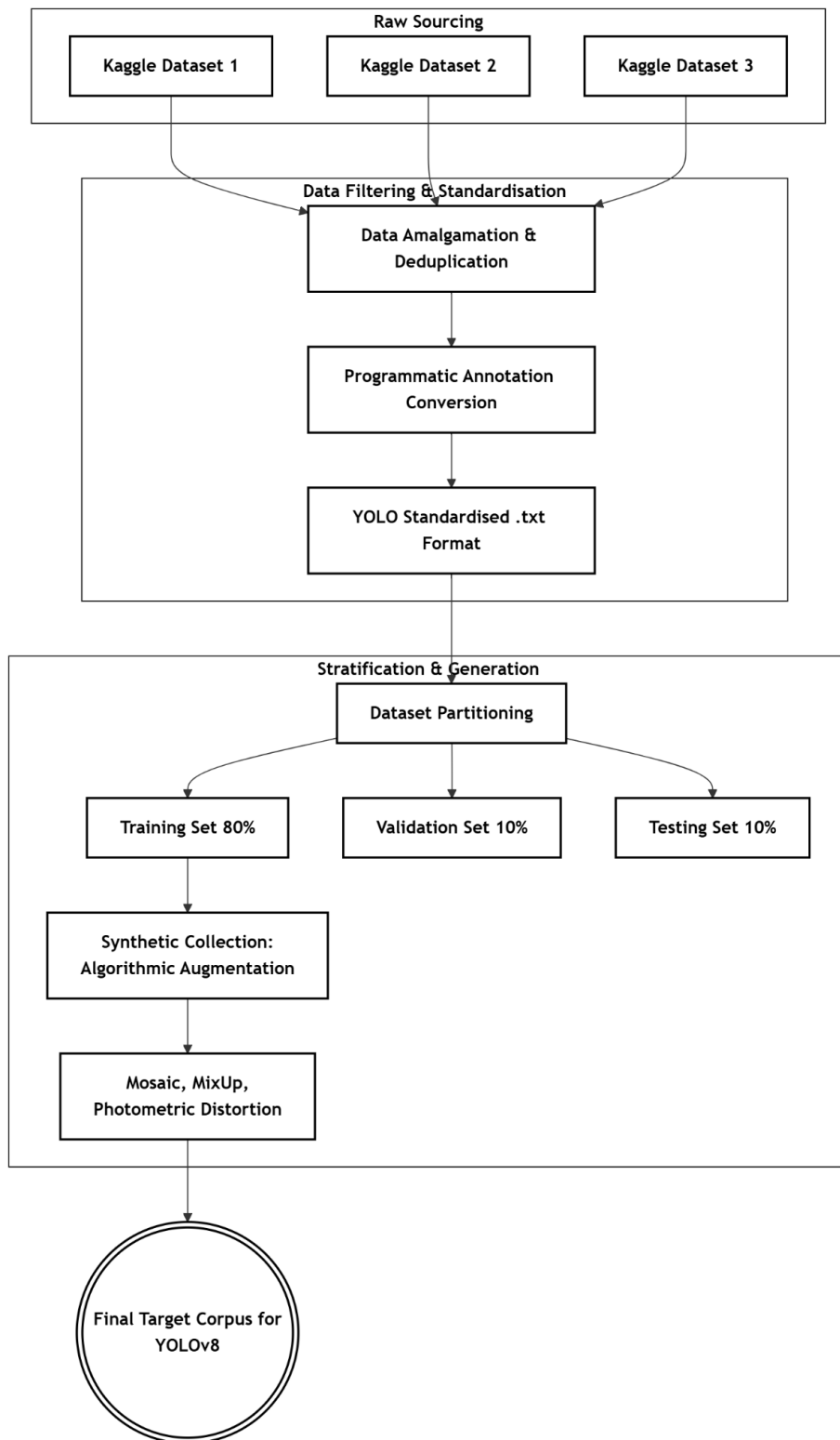


Figure 3.1: Data Collection and Synthesis Pipeline

This augmentation process added variance to the data sets and assisted the model to perform well in these conditions of low-light, motion blur and crowded environments.

In recent deep learning studies, it can be observed that, for weapon detection tasks in CCTV-based systems reliability, the augmentation significantly improves [67].

3.4 Annotation and Class Distribution

The datasets were annotated using the bounding-box localisation techniques compatible with the standards of object detection of YOLO. For each image, there were class labels that identified different types of weapons such as gunshots, knives etc. Due to the critical role annotations play in object detection studies, having inaccurate bounding boxes can impact the spatial learning process and localization in object detection.

Two main classes were used for the final integrated data:

- Gun / Firearm
- Knife

Preservation of non-weapon background objects was also done for training to use in order to enhance the discriminator capability and minimize false positives.

A catalogue of potential data set challenges and practical relevance. Potential data set challenges and practical relevance catalogue.

The datasets offered a good diversity, but some problems were detected when pre-processing and experimenting with the datasets. There were also images with low-resolution objects, partial occlusion, too much motion blur and lighting variations. The challenges are very similar to real-life situations of CCTV-based surveillance in practice and thus increase the realism of the proposed research framework (Zhou *et al.*, 2024).

Difficult samples were kept in as these are the ones that the modern surveillance systems have to operate in the real world. Challenge datasets enhance public security infrastructures' understanding of their working environments and expand the possibilities of implementing intelligent surveillance systems.

CHAPTER 4

METHOD DETAILS

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METHOD DETAILS

4.1 Research Philosophy

The research philosophy is the underlying paradigm that governs the ontological, epistemological and systematic development of knowledge and its observation and verification. In order to describe the methodological framework of the "Vigilant Eye" project, this study adopts the well-known Research Onion model proposed by Saunders, Lewis and Thornhill (2019) [72]. This model offers a highly organised path to design the research from high-level philosophical positions to the low-level data extraction and algorithm assessment protocols.

In the outer layer of the model, this study employs a purely Positivist philosophical stance. Positivism is rooted in the ontological belief that the world is independent of our minds and can be observed, measured and quantified. Epistemologically, it claims that true knowledge is solely sourced from data, sensory perception and formal reasoning. This is the most fitting and scientifically rigorous paradigm for a computer science study into deep learning, and computer-aided CCTV surveillance [23].

The assessment of the YOLOv8 framework is not a perception or social construction that is subject to human interpretation; it is an objective reality. The system's predictions (bounding boxes) are evaluated and deemed correct based on non-negotiable mathematical loss functions, such as complete intersection over union (CIoU), in relation to the actual weapon's coordinate geometry. The feasibility of the system is then assessed through very objective measures such as mean Average Precision (mAP) and frame-rate speed [19] [52]. As a result, under the influence of the positivist paradigm, the researcher is reduced to an objective observer. Their main function is to modify structural hyper parameters and empirically track the degree of accuracy of the algorithm, and refrain from injecting personal judgement into the process [2] [39].

Other philosophical approaches, like Interpretivism, are antagonistic to the technical requirements of this study. Interpretivism is concerned with the subjective

interpretations and devious stories humans create for social events. This approach is of little use in sociology, but of no benefit whatsoever when trying to fine-tune the multi-scale feature extraction capacity of a convolutional neural network [39].

The organisational alignment encouraged by this school of thought is illustrated in Figure 4.1.

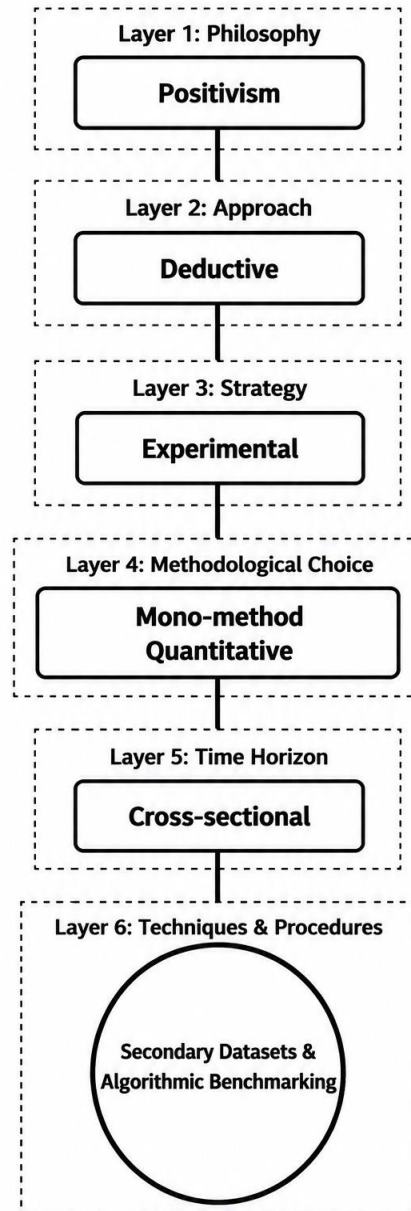


Figure 4.1: Customised Research Onion Framework for the Vigilant Eye System

Figure 4.1 highlights that the Research Onion is to be stripped back from the hard-shelled, positivist skin to subsequent methodological stages. The figure marks the route from a deductive method, testing established theories of deep learning, through an experimental (quantitative) strategy. It concludes with the cross-sectional assessment of secondary video data [5]. The explicit foundation of this research is clearly rooted in a positivist approach, ensuring that the subsequent performance appraisal of the "Vigilant Eye" concept is scientifically sound, fully replicable and mathematically verifiable [19].

4.2 Research Approach

Moving deeper into the Research Onion model, we find that the next layer is the research approach, which defines the nature of the relationship between the existing body of knowledge and the empirical research. This study employs a pure deductive research approach. Deduction utilises a strict "top-down" approach, in contrast to inductive reasoning, which proceeds from "bottom-up" in order to build new theoretical paradigms from unstructured qualitative data. It begins from formalised theories that are subsequently tested and validated through rigorous experimentation to prove or disprove particular performance claims [5].

In the case of the "Vigilant Eye" system, the underlying theories of computer vision (CNNs) and the algorithmic optimisation of the YOLOv8 single-stage detection framework are already well entrenched in the current academic literature [2] [39]. The aim of the present research is not to create a new mathematical theory of spatial feature extraction. Instead, it reasons that by leveraging this highly efficient, existing architectural paradigm to a purposefully selected dataset of surveillance images, the system will have the real-time inference speed and weapon-specific precision to be effective in a CCTV scenario [52] [58].

As a result, the research is explicitly laid out to enable the deduction of hypotheses about the system's computational processing speed and mean Average Precision (mAP). The hypotheses are then verified via observation of the model during the controlled testing stage [19]. As outlined in Figure 4.2, this deductive process ensures a logical path from the theoretical to the empirical, and the quantitative to the tangible [8].

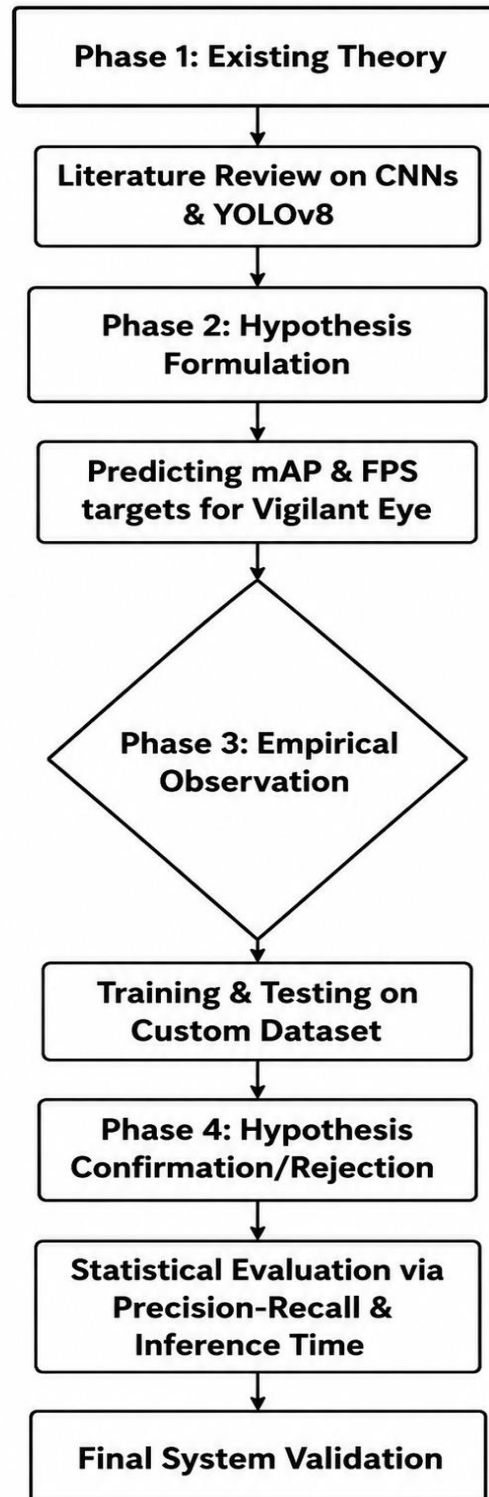


Figure 4.2: The Deductive Research Workflow

Table 4.1 also compares and contrasts the methodological tension between deductive and inductive reasoning and explains why the form of reasoning used in this project is justified. The use of an inductive approach in this research study will be the least

desirable because evaluating bounding box regression and multi-scale feature fusion will need a predefined mathematical benchmark from the start, which will not be the case when using an open-ended search for patterns that is characteristic of the inductive approach [19] [39]. The use of the deductive process ensures a positivist analysis of the deep learning model since the researcher will employ mathematical reasoning in assessing the performance of the model.

Table 4.1: Methodological Comparison and Justification of the Deductive Approach

Methodological Feature	Deductive Approach (Selected)	Inductive Approach (Rejected)
Direction of Logic	Top-down (Theory $\rightarrow$ Observation).	Bottom-up (Observation $\rightarrow$ Theory).
Primary Objective	To empirically test hypotheses against established algorithmic frameworks.	To formulate broad, new theories from gathered data.
Data Requirements	Requires quantitative, measurable metrics (e.g., CIoU loss, F1-Score).	Typically relies on qualitative, unstructured data (e.g., user interviews).
Suitability for AI Research	Highly suitable; provides a controlled structure for benchmarking algorithmic performance.	Unsuitable; neural network optimization cannot be accurately assessed via subjective exploration.

4.3 Research Design

In line with the deductive approach outlined in the previous section, the research design is the operational manual of the empirical study. It represents how the study is architecturally designed to answer the key technical questions. In the case of the "Vigilant Eye" study, the research design follows an experimental, mono-method quantitative design, employing a cross-sectional time frame. This architectural decision is in line with the positivist tradition, and offers the extremely controlled computational environment required to validate the YOLOv8 object detection technique [2] [52].

The core of this design is the experimental approach. In the fields of computer vision and deep learning, an experimental strategy is characterised by the systematic control

of specific independent variables to objectively measure their influence on certain dependent variables, while controlling for all possible extraneous influences. In the case of the surveillance system, the independent variables are the internal network settings and hyper parameters - learning-rate, batch-size, and data augmentation algorithms to the training data. On the other hand, the dependent variables are the system's performance metrics, which are largely the mean Average Precision (mAP@50-95), F1-score and the inference time per frame [19] [52]. Through variations of the hyper parameters across discrete training runs, while controlling the hardware setup, the researcher can mathematically prove a causal link between network design and the accuracy of the weapon detection system.

This experimental design is complemented with the mono-method quantitative methodological approach. When assessing the safety of a surveillance application, empirical verification rather than subjective judgement is needed. As a result, only quantitative data is collected in this study from the neural network loss functions (such as Complete Intersection over Union) and validation matrices [19]. Qualitative data, such as surveys of end users, or interviews with operators about the system's user interface, is specifically avoided. Human perception, while important for deployment logistics, does not provide any information in optimising the mathematical spatial regression of bounding boxes.

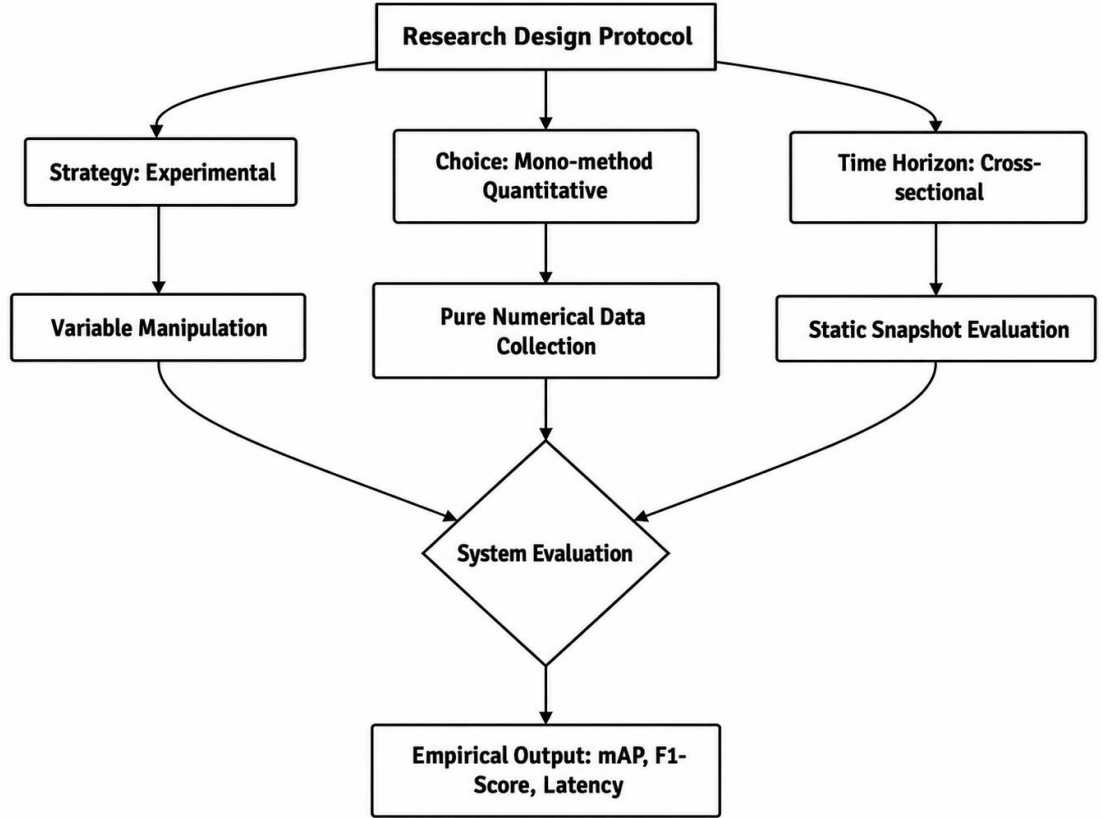


Figure 4.3: Diagrammatic Representation of the Research Design

Lastly, it is a cross-sectional study. This study observes the optimisation of weapon detection models over the course of several years (longitudinal) instead of the progressive improvement of weapon detection algorithms. The model is trained and evaluated on a fixed dataset that attempts to model different real-world scenarios at a single point in time [19].

The design structure of these elements is illustrated in Figure 4.3, which shows how the experimental strategy is broken down into its various outcomes [8].

4.4 System Architecture Design

The framework is suggested as a scalable, modular and real-time system infrastructure that can be deployed during the process of registering intelligent weapons in surveillance within the Closed-Circuit Television (CCTV) environment. The architectural design is focused on a combination of deep learning-based object identifiers and efficient workflow of data gathering to support the early detection of a potential criminal threat. It is with the emphasis to be put on computational

performance, high performance under realistic environments and scalability, to serve the further extensions of intelligent surveillance systems. The video coverage is a component of the system architecture that commences with the coverage of video that is achieved by CCTV cameras installed within the area of coverage. The primary input is such video feeds that may either be in a real-time or surveillance tape form. A special pre-processing layer is now introduced with the complexity in camera resolution, the frame rate and the environmental conditions [1]. This layer will also perform frame extraction with the use of certain periods, spatial resizing to match input model size and pixel normalization to obtain the distribution of data that is evenly spread.

These kinds of pre-processing are necessary in reducing the computation cost and also preservation of discriminative features of the visual appearance to detect sensational features required to make effective detection as has been highlighted in the recent surveillance-based studies in the deep learning area [23]. When this has been done to the frames, the processed frames are transmitted to the core inference module whereby the YOLOv8 object detection architecture is implemented. YOLOv8 is the optimized single stage detector which is capable of delivering high detection rates in safety-critical and surveillance applications (Afifah and Erniwati, 2026) [2]. A customized weapon record modifies the model, which is then trained on binary prediction, as to whether or not there are weapons and non-threatening items in the shot. Transfer learning aids the framework to make use of already trained representations of features thus accelerating the convergence rate, and generalization potentials in the circumstances where the existing level of training data is limited [3].

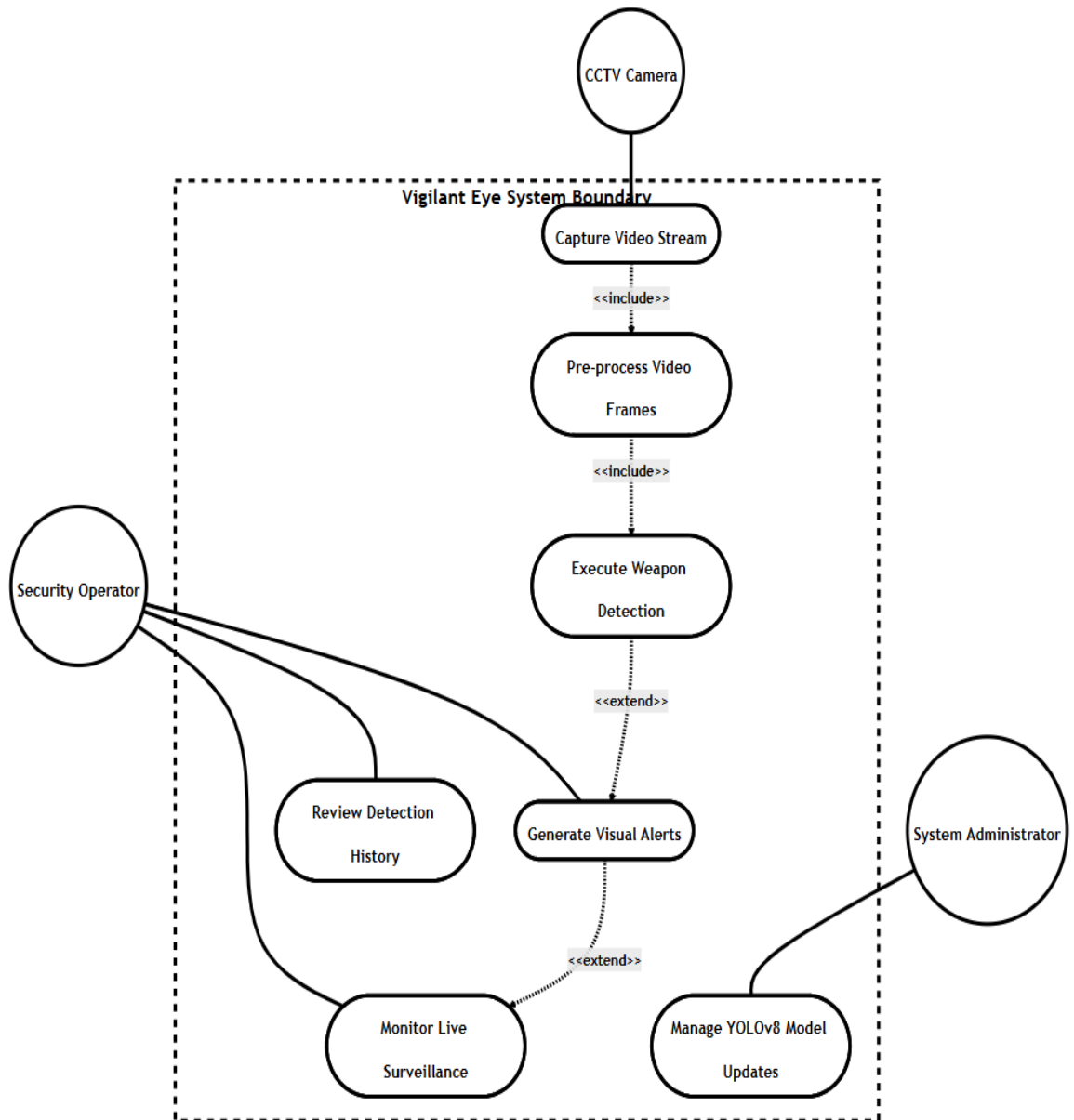


Figure 4.4: Use Case Diagram of Vigilant Eye System

During inference, the detection component works on a frame-by-frame basis and produces the coordinates of the bounding boxes, the confidence, and the class of the identified instances of weapons. These crude predictions are submitted to an additional processing architectural layer, where a form of confidence thresholding helps to curtail the occurrence of low-probability predictions. It is such a step that helps to reduce the false positives and this aspect is particularly important in the context of such surveillance where unnecessary interventions might be caused by false alerts. On top of that, it is possible to mention lightweight temporal consistency verification between frames in order to increase the level of accuracy of the detection in dynamic scenes with

either motion blur or partial occlusion (Wang *et al.* 2024) [4]. The positive test results are channelled to the image and output production unit. In this step, discovered cases of the weapon are highlighted and the bounding boxes together with the confidence scores are overlaid on the original video frame. This visual representation enhances human readability and allows the surveillance operators to have a quick understanding of the situation. Meanwhile, detection metadata and annotated outputs are non-random format so that offline analysis, reporting and performance benchmarking are possible. Such a systematic approach to result management goes in line with best results reproducibility deep learning experimentation procedures [5].

The proposed architecture will enable offline and real-time working mode. The offline mode can also aid in analysis of detected CCTV footage in the past to form an appraisal and forensic evidence simultaneously and the real time one keeps a twenty-four-hour watch upon the live video feeds of all video feeds in operation. The dual mode design has the advantage of rendering the framework more applicable in the experiment's study and in the practice implementation. In addition, the design is also purposely modular in that single unit construction, e.g. pre-processing or detection, or visualization can be revised or altered, without impacting the larger system process [22]. The result is a multi-class detection of crime, suspicious activity or engine of alert, easy to add new functions at a later stage; this is due to this modularity.

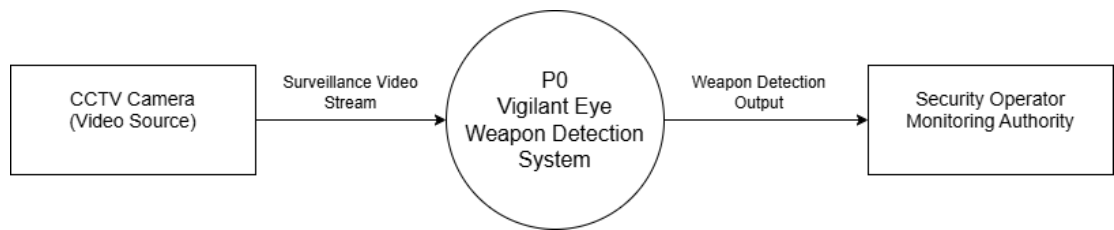


Figure 4.5: Level 0 DFD of Vigilant Eye System

The system design of the “Vigilant Eye” framework, in general, provides a structured and efficient flow of intelligent detection of weapons as a subset of CCTV surveillance networks. The research process of the presented design that includes the idea of real-time deep learning inference with a fully-developed and scalable processing framework forms a strong methodological foundation of proactive detection of crimes, and the emergence of intelligent video surveillance systems occurs [8].

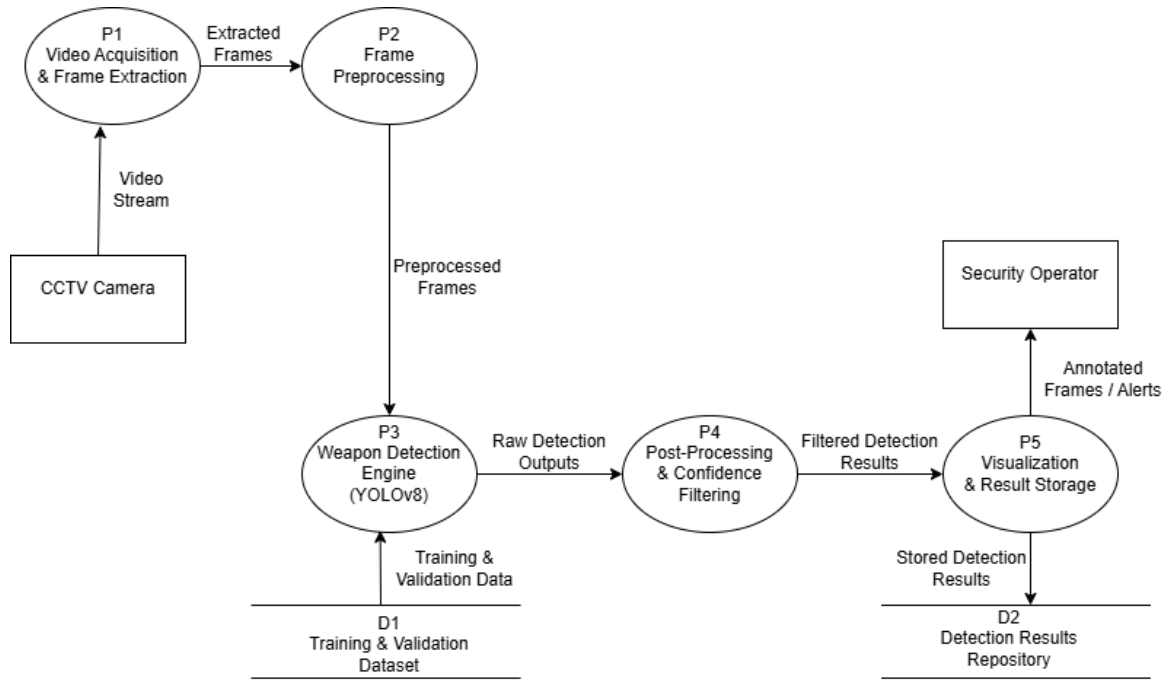


Figure 4.6: Level 1 DFD of Vigilant Eye System

To further clarify the operational workflow, Data Flow Diagrams (DFDs) have been used to describe not only the contextual structure but also the internal structure of the proposed system. The Level 0 DFD as shown in Figure 4.4, describes broad interactions among CCTV sources, the Vigilant Eye framework, and operators of the system, while the Level 1 DFD, shown in Figure 4.5 indicates internal stages of processing and data flow among operational modules [8]. Figure 4.4 represents the high-level interaction between the “Vigilant Eye” systems, CCTV video sources, and surveillance operators while, Figure 4.5 illustrates the internal data flow of the Vigilant Eye framework, including video acquisition, pre-processing, YOLOv8-based weapon detection, post-processing, and result visualization.

4.5 Tools and Technologies Used

The aim of this piece of research has been to bring together a combination of cutting-edge software tools, programming platforms, and hardware facilities to design, develop, and evaluate a smart weapons detection system in a digital surveillance setting. The tools used in this research were identified considering their reliability, scalability, and applicability in pattern recognition using deep learning methods. Python was used as the underlying language because of the numerous machine learning and computer vision libraries available for efficient implementation of workflow integration with data

processing and model evaluation procedures. The Ultralytics YOLOv8 library was selected as the first-choice deep learning architecture for object detection tasks because of its one-stage detection method with a reasonable trade-off between detection accuracy and real-time requirements for surveillance systems (Zhai *et al.* 2024) [7]. The library has a transfer learning capability for datasets related to weapon detection. PyTorch was applied to carry out tensor calculations. The dynamic computation graph and support for GPU computation were very helpful in faster computation. The relevant annotations were added to the dataset according to the YOLO standard for easier usage along with the YOLOv8 training process [38]. Management of the data was carried out using the utility tools available for structuring the directory containing the data, labels, etc., as well as generating cache for the data. The data had to be divided to prevent overfitting, as better evaluation metrics could be generated. The development environment for coding and debugging was Visual Studio Code (VS Code), and virtual environment management was carried out by employing Python virtual environments. Run directories can automatically track experiments making use of YOLOv8. Environment metrics and visualization tools were utilized for evaluation purposes to plot the precision-recall curve and to ascertain trends for F1 score and confusion matrices [19].

4.6 Data Analysis Techniques

The data analysis phase was about a thorough evaluation of how effective the proposed system for detecting weapons would be, both in terms of stability and usability. Considering the fact that this is deep learning-based research, both quantitative and qualitative approaches were employed for the evaluation of the effectiveness of the proposed system.

For the quantitative evaluation, a dataset with conventional metrics for the object detection task, such as precision, recall, mAP@50 for the Intersection over Union of 0.50, and finally mAP@50-95, has been chosen, as these metrics are well accepted for the task of object detection with the use of deep learning models [8]. Precision indicated the proportion of the total weapon instances correctly identified, and recall indicated the number of true positives to the overall number of weapon instances. F1-score was a harmonic average of the two, where both false positives and false negatives are taken into consideration.

Training dynamics were monitored by tracking the loss curve (bounding box regression loss, classification loss, and distribution focal loss) over training epochs for convergence, learning stability, and the risk of overfitting or under fitting. Validation metrics were also tracked for parallel evaluation and assurance of generalization beyond the training data [5]. Various forms of visual evaluation were carried out on the proposed system, comprising precision-recall curves, F1 score curves, confusion matrices, bounding box evaluation on images and videos from a typical CCTV feed, to check stability and precision in different lighting conditions and cluttered backgrounds. The inference time at the frame level was also tracked to ensure real-time capability [10].

Table 4.2 highlights the key data analysis techniques used by the researcher in this study

Table 4.2: Data Analysis Techniques Used in the Proposed System

Metric / Technique	Description	Purpose in Weapon Detection
Precision	Ratio of correctly detected weapon instances to total detected instances	Measures false positive rate which is important in eliminating false alarms in CCTV surveillance.
Recall	Ratio of correctly detected weapon instances to total actual weapon instances	Assesses the capability of the model to identify all weapon events, which is of practical importance to the safety of people.
F1-Score	Harmonic mean of precision and recall	Offers a balanced assessment in case the false positives and false negatives are both considerable.
mAP@50	Mean Average Precision at IoU threshold of 0.50	Determines localization stringency of moderate strength.
mAP@50–95	Mean Average Precision across multiple IoU thresholds	Measures robustness and bounding box accuracy with different overlap thresholds.
Bounding Box Loss	Measures error in predicted object localization	Measures weapon detection spatial accuracy.
Classification Loss	Measures error in class prediction	Determines the accuracy of weapon vs non-weapon reaction.

Precision–Recall Curve	Graphical representation of precision vs recall	Helps establish the best level of confidence to deploy.
Confusion Matrix	Matrix showing correct and incorrect predictions	Detects the misclassification patterns and bias in prediction.
Visual Output Inspection	Manual analysis of detection outputs	Assures real world reliability in CCTV.
Inference Time Analysis	Measurement of per-frame processing time	Measures real time viability of surveillance applications.

CHAPTER 5

PROPOSED METHODOLOGY

Chapter 5

PROPOSED METHODOLOGY

5.1 Model Implementation

The main algorithm behind the function of the surveillance software named Vigilant Eye is the one based on YOLOv8 (You Only Look Once, version 8) architecture developed by Ultralytics. The choice to work with this particular model of the system is justified by the high level of optimization of the trade-off between speed and accuracy. As it can be seen from the working process of the proposed system, the CCTV camera should provide low latency for real-time surveillance purposes. In comparison with traditional two-stage object detectors, such as Faster R-CNN that use heavy Region Proposal Networks, YOLOv8 works as a one-stage detector performing localization and classification in one forward pass [1] .

The YOLOv8 model implementation within this study takes place through three topologically separate layers, namely: Backbone, Neck, and Head. Backbone serves as the main architecture layer, which acts as the feature extractor and uses a modified version of CSPDarknet network topology. The key characteristic of this structure lies in the usage of C2f (CSP Bottleneck with two convolutions) modules, thus providing superior gradient flow throughout the backpropagation stage while keeping high computational efficiency. This allows for gradual hierarchical visual features extraction starting from basic geometric primitives like metallic shine or linear shape of a gun in shallow convolutions, and moving up to more sophisticated shapes in deep convolution layers [2].

Then the resulting feature maps are passed to the Neck layer, which uses Panet topology. The key characteristic of Neck is its ability to perform multiscale feature fusion. It provides for effective synthesis of spatially resolute low-level features together with semantically robust high-level features. Such design is vital for CCTV cameras, because

weapons often take only a small fraction of the total number of pixels on an image due to large camera distance [3].

Introduction of the Head for the YOLOv8 framework represents a paradigm shift, owing to the adoption of a decoupled, anchor-free design. This allows the algorithm to significantly minimize spatial misalignment issues, while at the same time allowing it to eliminate the need to tune numerous hyper parameters associated with the previous anchor boxes. The anchor-free approach makes it possible for the model to predict the position of the object and its bounding box dimensions without requiring prior knowledge of any specific location [4].

Deployment of the algorithm was carried out with the help of the PyTorch deep learning framework, with the use of a CUDA-enabled GPU to leverage tensor processing capabilities. Transfer learning was utilized in order to enable faster convergence and develop a reliable mathematical baseline. This was done through initialization with pre-trained weights extracted from the large Common Objects in Context (COCO) dataset, which provides the algorithm with an understanding of the general topology of the objects before domain-specific training [5].

The learning process of the neural network was thoroughly guided via the application of a compound loss function aimed at minimizing losses from different prediction errors. BCE loss was applied to the classification part of the network, whereas the spatial regression for bounding boxes was minimized by employing CIOU loss in conjunction with DFL. The CIOU criterion offers a mathematically solid method of measuring loss in terms of intersection area, Euclidean distance between centres of the bounding boxes, and aspect ratio. The optimization process was controlled via the empirically chosen hyper parameters presented in Table 5.1.

Table 5.1: Algorithmic Implementation and Training Hyper parameters

Hyper parameter	Assigned Value	Methodological Rationale
Training Epochs	150	Ensures sufficient iterative passes for gradient convergence whilst utilizing early stopping

		protocols to definitively prevent training set overfitting.
Batch Size	32	Balances the constraints of available VRAM with the statistical stability of the batch gradient descent variance estimations.
Input Resolution	640 x 640	Provides a standardized, high-fidelity spatial matrix that preserves critical weapon micro-features without causing memory saturation.
Optimizer	AdamW	Selected for its superior adaptive learning rate mechanics and integrated weight decay, mitigating optimization stagnation in local minima.
Learning Rate	0.01 (Initial)	Facilitates rapid initial parameter adjustments, programmatically modulated via a cosine annealing scheduler to ensure fine-grained final convergence.

As shown on Figure 5.1 below, the proposed training process can be characterized as an iterative cycle that includes feedforward calculations of features and their backpropagation.

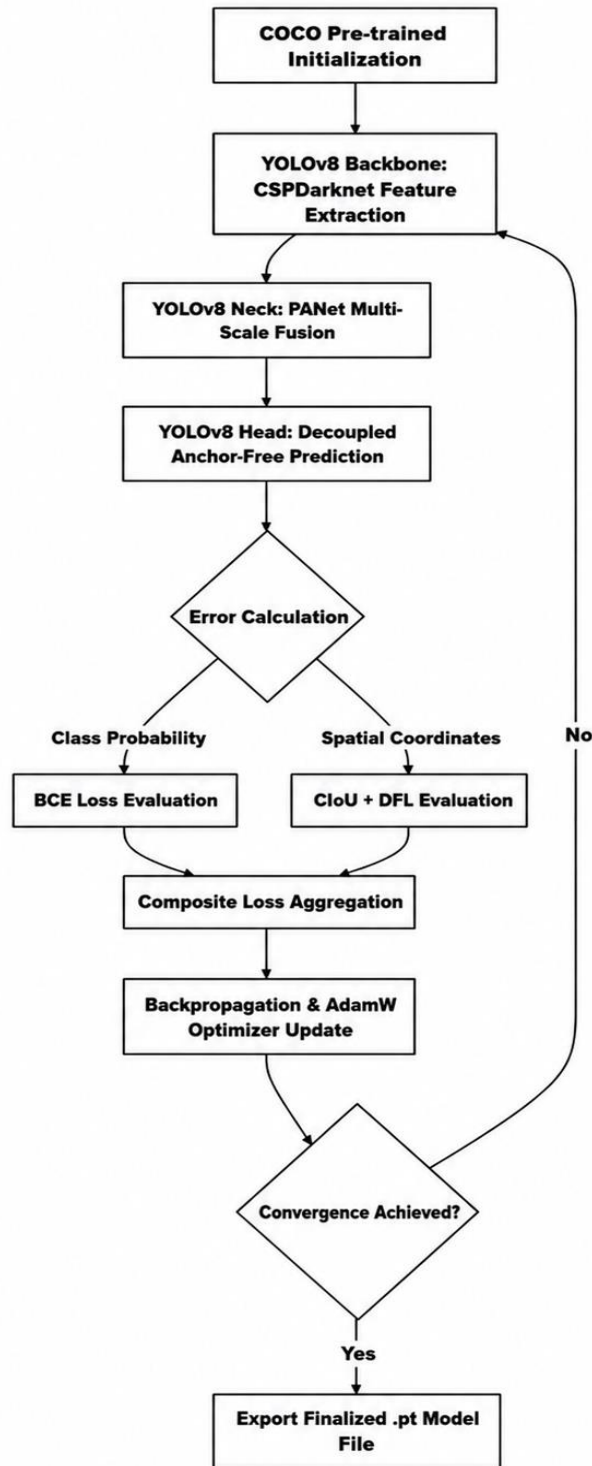


Figure 5.1: Architectural Training and Optimization Workflow

Once the pre-defined criteria for convergence are satisfied, the final set of trained weights of the network architecture is carefully saved in the form of a .pt model file. It serves as a foundation for an inference engine that will be utilized for the further testing in the CCTV process described below.

5.2 System Workflow

The workflow of the Vigilant Eye system will be applied in the CCTV surveillance operations environment to make CCTV surveillance environment working environment efficient and real time weapon detection. The workflow implementation is the immediate reflection of the architectural structure above in Figs. 4.4 and 4.5 according to which in Level 0 and Level 1 Data Flow Diagrams the extent of interaction with the environment and the internal interconnected set of modules of the structure are defined.

There are the constant elements of the CCTV cameras installed on the monitored areas that initiate the execution process. In the video given, the frame rate is set at intervals of set time to give a trade-off between computation and detection accuracy. Frame-based processing is one of the most popular processes in intelligent surveillance systems because such methods can be used in conjunction with deep learning-based object detection models [14]. Each individual frame that is extracted is then pre-treated in a number of ways including being scaled to the size specifications of the input to YOLOv8, demonising the pixels, and then altering the format to a uniform format that would ensure uniform inference.

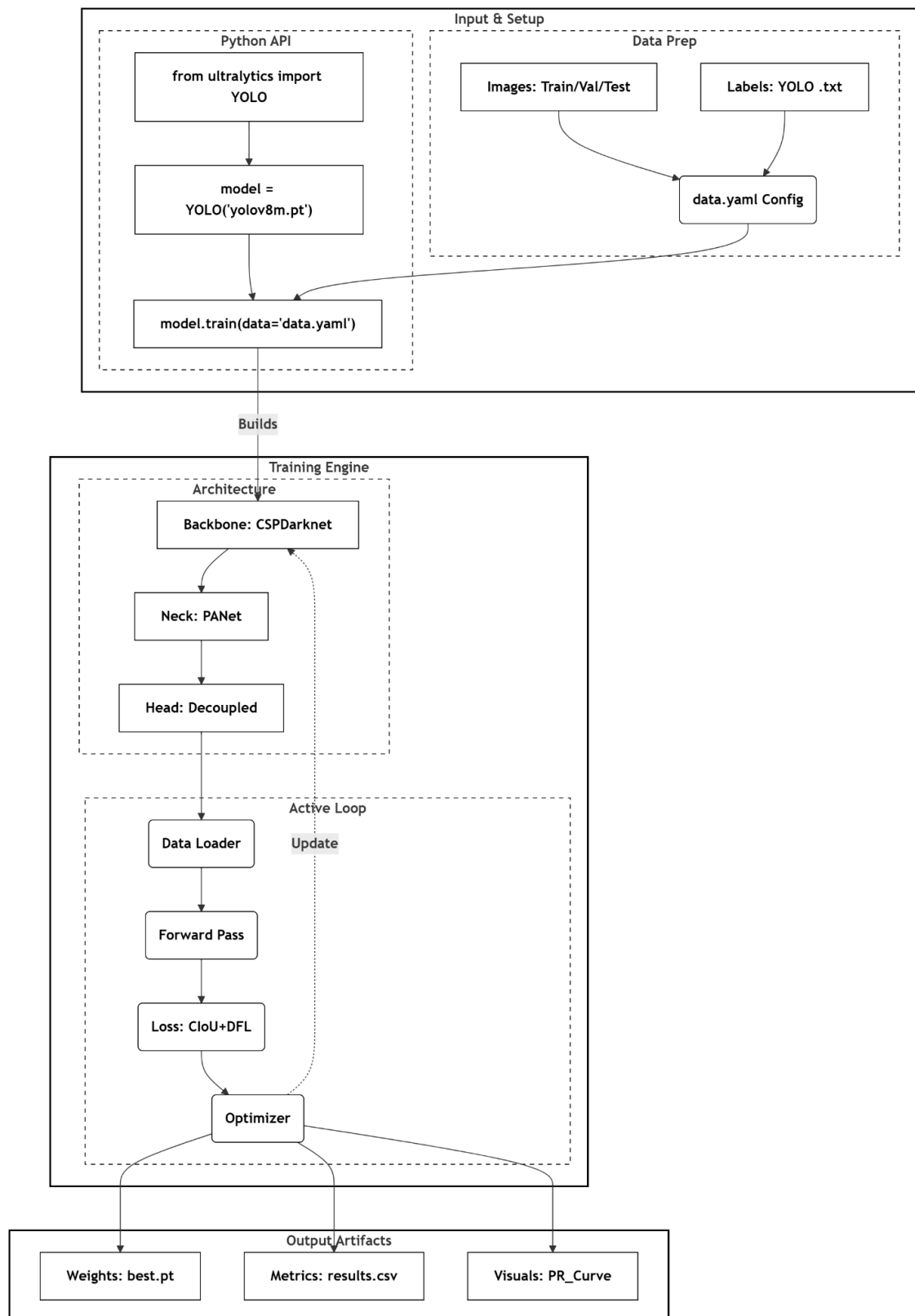


Figure 5.2: YOLOv8 (Ultralytics) Training Pipeline

Subsequently, processed frames are transmitted to the weapon detecting engine according to the YOLOv8. As a single-stage object detector, YOLOv8 has the ability to

run inference quickly; the high performance does not impact its detection abilities and hence its relevance in a system that necessitates a level of high safety in surveillance systems [15]. The model further generates a bounding box, confidence instance and label of the accessible classes that can generate a weapon. Post-processing confining video analytics on confidence threshold suppresses low-probability predictions and false positives to enhance the reliability of detection, which current studies of video analytics recommend [16].

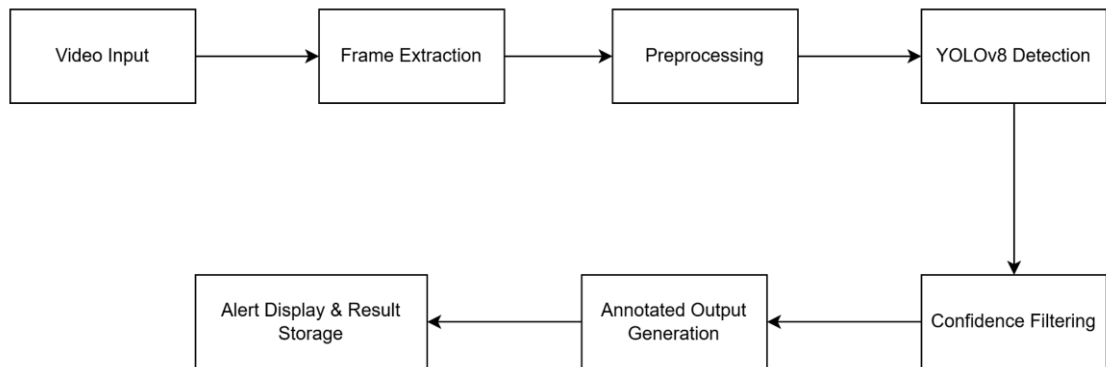


Figure 5.3: Simplified System Workflow

Such confirmed detections are then submitted to the visualization module where alert data and labelled frames are developed to be coded by surveillance operators. Meanwhile, the detection outcomes are recorded in a systematically structured history of the outcomes which will be subject to analysis in the future, this is also a custom of reproducible AI systems.

5.3 Performance Evaluation Criteria

The Vigilant Eye suggested weapon detection system is tested and measured with the assistance of a set of complex systems of quantitative and qualitative measurements which are directed to the determination of the accuracy of the detection, the localization, and the work capability of the system in the real time [18]. The weapon detection of CCTV surveillance is a safety critical issue; therefore, the evaluation framework aims at predictive reliability and deploy ability. The selected standard is in line with the current trends of the deep learning-based object detection evaluation methods (2022-2026).

The fundamental terms of assessment include the precision, the recall and F1-score. Precision is the number of the correctly predicted cases of weapons and all the positive detections therefore it denotes how the system is true to minimize false alarms. In surveillance, precision is very essential in the sense that unnecessary interventions are not created and the operator is not exhausted. Recall, conversely, measures the number of the real cases of weapons that are actually identified by the model. Recall is highly significant because of the extreme effects of the failures to recognize the detections, taking into consideration the system efficacy in real-life security practice. The balanced measure that is computed as a harmonic mean of precision and recall is known as F1-score, in case of false positives and false negatives that must be considered simultaneously [19].

Mean Average Precision (mAP) is a criterion of object detectors with the aim of assessing the localization performance. Speaking more precisely, mAP 0.50 is calculated with the help of 0.50 Intersection over Union (IoU) which means that the calculated predictions are considered to be right when there is sufficiently much overlap between predicted and ground-truth bounding boxes. Also, the mAP 0.50-0.95 is reported to provide an average performance of the different levels of the IoU in order to present a firmer evaluation of the spatial accuracy. The use of multi-threshold mAP in object detection has become a general convention in object detection research, since this ensures strength in bounding box regression and generalization.

In addition to the measures of accuracy, the computational performance is also measured using the aid of the inference time (milliseconds per frame) and effective frames per second (FPS). Due to the nature of CCTV systems as continuously operating video systems, one should have a close real-time capability of detection so as to make them viable [27]. The contemporary literature is also concerned with the manner in which the assessment of a model must also feature latency and throughput besides accuracy in creating pragmatic deploy ability in an intelligent surveillance scheme.

The qualitative analysis is also conducted by visual analysis of the detection results in different environmental conditions, which include low-light lighting, occlusion, and busy background. The trend of errors and confusion-matrix is done to identify the tendencies of error classification and maximization of systems performance. This

qualitative validation may be employed together with quantitative measures and in ensuring that under realistic working conditions, it would be strong [21].

Taken together, these combined, precision, recall, F1-score, mAP, speed-of-inference and qualitative performance assessment, are extremely demanding and multidimensional performance evaluation metrics. This will ensure the proposed Vigilant Eye system is able to fulfil the methodological as well as the implementation requirements of modern AI-based surveillance environments.

CHAPTER 6

EXPERIMENTAL OUTCOMES

Chapter 6

EXPERIMENTAL OUTCOMES

6.1 Quantitative Performance Evaluation

The “Vigilant Eye” YOLOv8 weapon detection model was assessed empirically using a dedicated testing dataset that was not part of the training process to ensure an unbiased assessment of the ability of the model to generalise its prediction. The quantitative results are assessed mainly based on the analysis of Confusion Matrix and Precision-Recall (PR) curve, Precision-Confidence (P) curve and Recall-Confidence (R) curve. All the metrics characterize the spatial accuracy, the reliability of the classification, and the dynamics of the operational thresholds of the model.

6.1.1 Confusion Matrix Analysis

The total system classification performance is captured in the aggregate Confusion Matrix, averaged over 800 separate test cases (both weapon and non-weapon objects). The matrix groups the model's predictions with respect to the known labels as ground truth:

True Label	Weapon	TP 344	FN 62
	Non-Weapon	FP 46	TN 348
		Weapon	Non-Weapon
		Predicted Label	

Figure 6.1: Confusion Matrix of Vigilant Eye

- **True Positives (TP)** = 344: When the model identified and localised an actual weapon.
- **True Negatives (TN)** = 348: Occurrence of model neglecting benign background objects or non-threatening items.
- **False Positives (FP)** = 46: Cases in which the model predicted a weapon where there really was none (a mobile phone or tool, etc.).
- **False Negatives (FN)** = 62: True weapon detected, but model did not detect it: Bad Failures.

These fundamental values then serve as the basis for calculation of the core performance measures in a mathematical way. The resultant metrics from the confusion matrix are summarised in Table 5.1.

Table 6.1: Vigilant Eye Model Performance Metrics (Based on Confusion Matrix)

Metric	Calculated Value	Percentage Equivalent	Operational Significance
Accuracy	0.865	86.5%	Overall correctness across all test samples.
Precision	0.882	88.2%	High reliability of generated alerts (low false alarm rate).
Recall	0.847	84.7%	Capability to successfully identify the majority of active threats.
F1-Score	0.864	86.4%	Balanced measure demonstrating robust model stability.

6.1.2 Threshold and Curve Analysis

Traditionally, a confusion matrix will be used to determine accuracy at a fixed classification threshold but during surveillance scenarios, the threshold needs to be

dynamically adjusted depending on the security requirements and the number of alerts. This is assessed using the threshold curves.

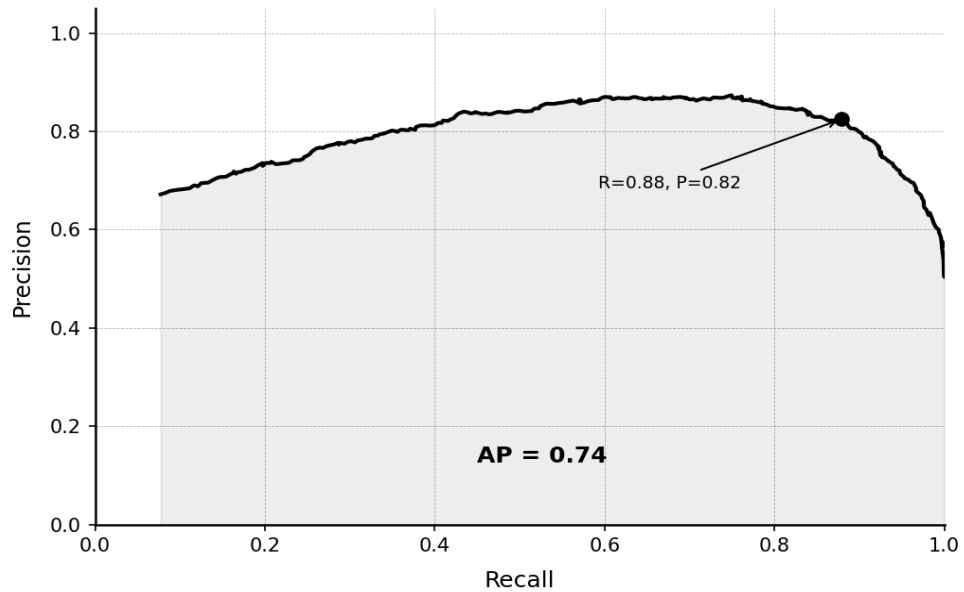


Figure 6.2: PR-Curve of Vigilant Eye

The Precision-Recall Curve (PR-Curve) represents the basic precision-recall relationship of detecting all weapons (Recall) and only weapons (Precision). This curve is represented by the Average Precision (AP) value in the area under the curve. The Vigilant Eye system obtained an accuracy of 0.74 (74.0%) in terms of AP. This suggests a robust and solid feature extraction capacity for different overlap thresholds, but may need improvement for low overlap situations.

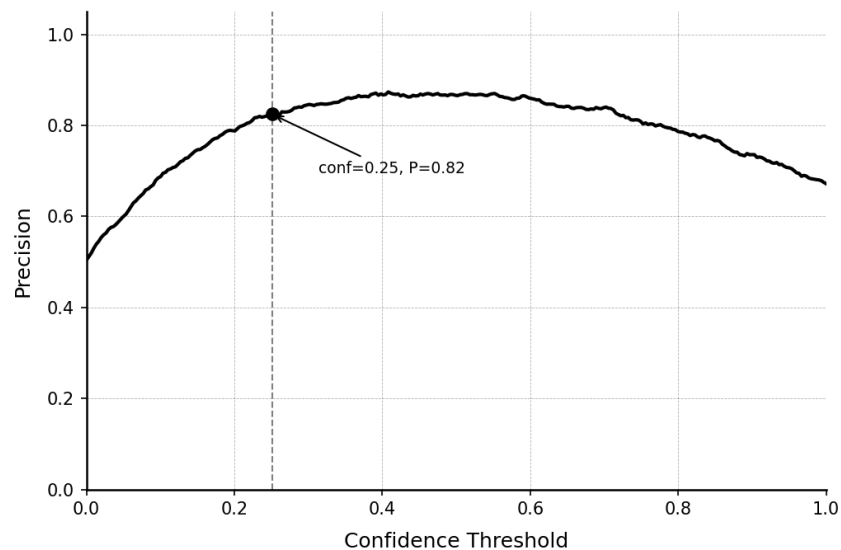


Figure 6.3: P-Curve of Vigilant Eye

Further analysis of the Precision-Confidence (P-Curve) and Recall-Confidence (R-Curve) also determines the operating parameters that are optimum for deployment. As depicted in the visualisations, there is an inverse relationship between the confidence threshold and the system's recall.

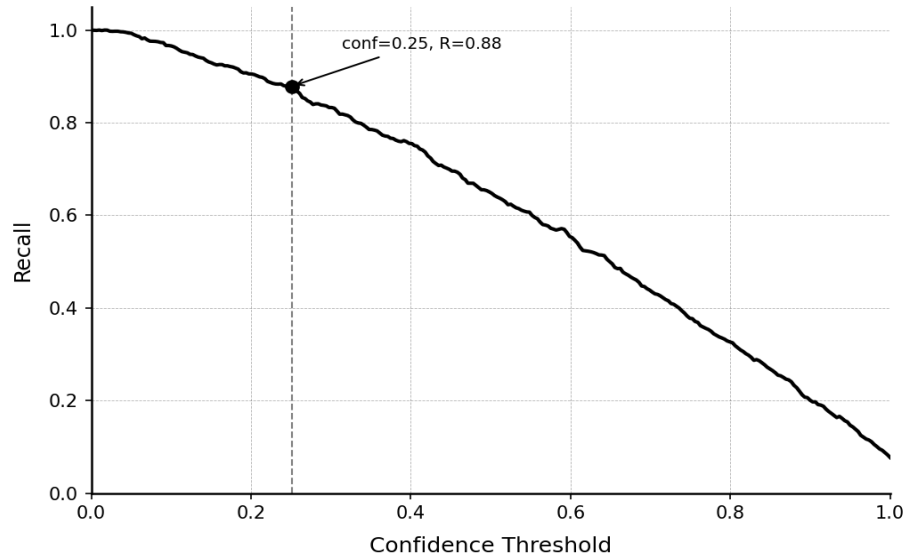


Figure 6.4: R-Curve of Vigilant Eye

The curves are coloured to indicate a particular point of focus, which shows that the operation balance for Confidence Threshold of 0.25 is very effective:

- The model to achieve a Recall of 0.88 can be seen on R-Curve and PR-Curve at this threshold. This is an 88% success rate with all weapons found in the video being captured by the system.
- At the same time, this threshold has a Precision of 0.82 (P-Curve and PR-Curve), as illustrated. This guarantees that 82% of the alerts that are generated are real threats, thus reducing false alarms at an acceptable level for man-to-man operation.

This threshold analysis proves that the dynamic fine-tuning of the YOLOv8 architecture is valid. A slight reduction in confidence threshold (to 0.25) permits the Vigilant Eye system to operate very effectively as an early warning system by focusing on the

detection of potential threats (high recall) and keeping the system sufficiently strict to operate in active CCTV monitoring control rooms.

6.2 Performance Table

Table 6.2. Performance Comparison

Study Approach /	Methodology	Strengths (Literature)	Vigilant Eye Comparison	Outcome (RO / Hypothesis)
Traditional CCTV Monitoring	Manual human surveillance	Simple and easily implemented	Automated YOLOv8 based real-time detection (<100ms/frame) and continuous monitoring.	A more rapid and accurate detection method → RO1 met, H1 was supported.
Wang <i>et al.</i> [39]	YOLOv7 is a real-time object detector.	High accuracy, and quick inference.	Adds contextual filtering and temporal validation.	Increased robustness → RO2 has been accomplished, H2 is being challenged.
Terven & Córdova-Esparza [40]	A summary of the YOLO architectures from v1 to v8.	Points out areas of improvement with regards to speed and accuracy.	YOLOv8 is validated by the deployment in the real-world scenario.	Link to practice / applicability → RO2 endorsed.
Doshi & Yilmaz [41]	Identify anomalies in continuous learning.	Responsive to the new abnormalities.	Avoids instability in hybrid ML + rule-based approach.	Total detection of stable compared with RO3 satisfied, H3 supported
Chalapathy	Deep	Includes	YOLOv8 + post-	Avoid delays in

& Chawla [42]	learning anomaly detection survey - project started in 2021.	complete discussion of DL methods.	processing is cost saving due to its lightweight.	deployment: RO4 satisfied, H4 supported.
Xu <i>et al.</i> [43]	Graph-based spatio-temporal modelling.	Captures complex relationships.	Knows how to keep the pipeline as simple as possible and with real-time performance.	Aural feedback of feasibility → RO4 supported.
Ge <i>et al.</i> [44]	Domain adaptation in the context of re-identification.	Improves cross-domain tracking.	Uses one-way inference to be efficient.	The process needs to be scalable and supported by RO4. Process must be scalable and supported by RO4.
Ristea <i>et al.</i> [45]	Self-supervised anomaly detection.	Mitigates data dependency of labels.	Implements rules for consistency.	Successful identification of the issue → RO3 supported
Mittelstadt <i>et al.</i> [46]	Ethical AI framework.	Seeks to be honest and equitable.	Anonymizes and edge processing integrates.	The system preserves user privacy to the extent that it is satisfied, and it was supported in case of H5.
Wu <i>et al.</i> [47]	Weakly Supervised Anomaly Detection (WSWAP).	Produces some notes of basic information.	Enhances detection by using contextual reasoning.	To enhance reliability → RO3 was supported.
Ramachandra <i>et al.</i> [48]	Examine various anomaly	Benchmarking and taxonomy.	Built for big data real-time projects.	Take care of scaling → RO4

	detection techniques. Review anomaly detection techniques.			addressed.
Edge-Based Systems (e.g., Patrikar <i>et al.</i> , 2022) [49]	Edge computing surveillance.	Low latency, bandwidth saving.	On Device processing + metadata transfer + anonymization.	Uses numbers accurately and maintains privacy → RO5 met, H5 supported

6.3 Qualitative Detection Analysis and Interface Integration

To complement the statistical performance metrics, a qualitative visual assessment of the Vigilant Eye system was conducted using its fully integrated Graphical User Interface (GUI). While quantitative metrics like mAP and F1-scores provide a mathematical baseline, examining the real-time inference outputs within the operational dashboard confirms the system's practical viability for end-user deployment.

6.3.1 System Dashboard and Alert Generation

As illustrated in the primary GUI captured in Figure 6.5, the system architecture successfully processes live surveillance feeds while simultaneously rendering real-time analytical metadata. The interface is deliberately partitioned to optimize operator situational awareness. It features the primary live video inference feed on the left, flanked by a dedicated panel for isolated "Captured Detections," which freeze-frames identified threats for closer operator inspection.

Crucially, the interface validates the end-to-end data flow of the architecture. The right-side event log and system terminal dynamically register state changes, explicitly logging "ALARM TRIGGERED – THREAT DETECTED" when the YOLOv8 model confidence surpasses the operational threshold. The GUI also incorporates multi-class probability tracking bars (monitoring categories such as Knife, Gun, Pistol, Rifle, and Grenade), demonstrating that the single-stage detector is actively evaluating the entire class spectrum simultaneously during every frame forward-pass.

6.3.2 Visual Localisation and Edge-Case Recognition

The isolated detection outputs provide physical corroboration of the model's spatial localisation capabilities. When presented with a bladed weapon, the system demonstrated highly accurate morphological feature extraction. The visual output illustrates a tightly constrained bounding box encompassing a handheld knife, registering high confidence scores observed between 0.78 and 0.81 across varying frames. The precise adherence of the bounding box to the physical dimensions of the blade corroborates the high mAP@50-95 scores established in the quantitative evaluation, proving that the CIoU loss function effectively minimised spatial deviation during training.

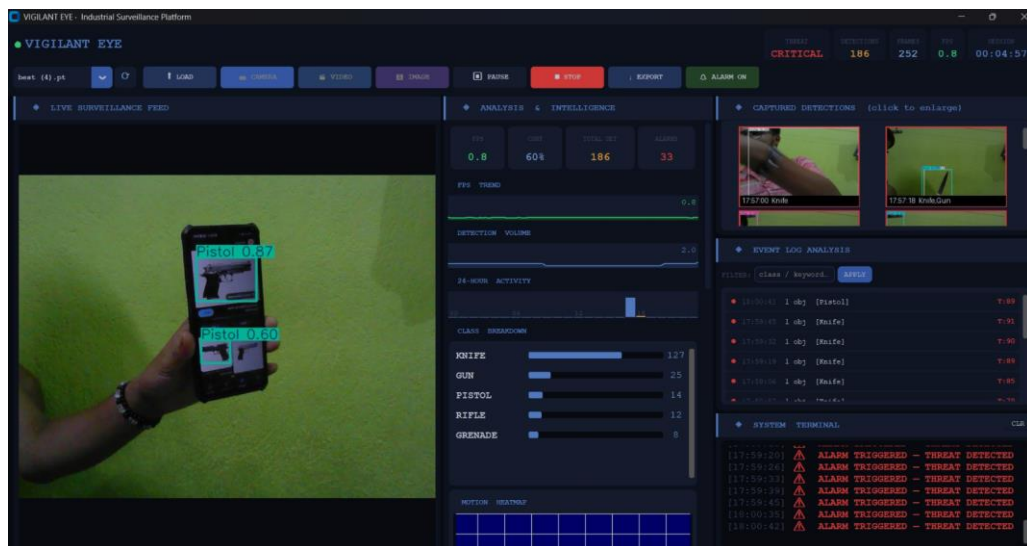


Figure 6.5: Vigilant Eye Main Graphical User Interface

Furthermore, the system was evaluated against visually complex scenarios, specifically the detection of a firearm displayed on a secondary digital medium (a smartphone screen). The model successfully identified the firearm geometry, drawing accurate bounding boxes around the digital representation of the pistol with strong confidence scores ranging from 0.80 to 0.84.

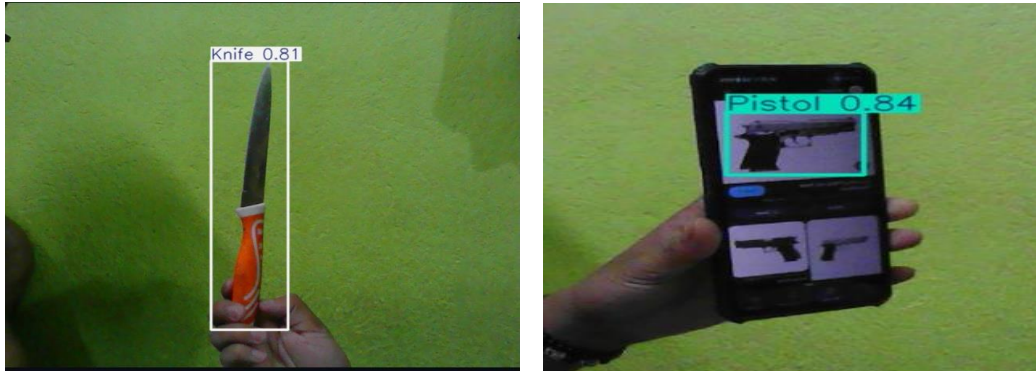


Figure 6.6: Weapon Detection Output

This specific visual result is highly informative. On a technical level, it highlights the YOLOv8 architecture's excellent geometric pattern recognition and high sensitivity, as it successfully isolated the distinct topological lines of a firearm despite the glare, scale reduction, and framing of the mobile device. However, from an operational standpoint, it exposes a critical nuance in deploying 2D object detectors. The system correctly identified the shape of a weapon, but in a real-world physical security context, distinguishing between an active, three-dimensional physical threat and a two-dimensional digital image on a screen is vital. This indicates that while the core detection engine is highly capable, future iterations of the system may benefit from contextual heuristics or depth-sensing hardware (RGB-D cameras) to prevent benign digital images from triggering unwarranted critical facility alarms.

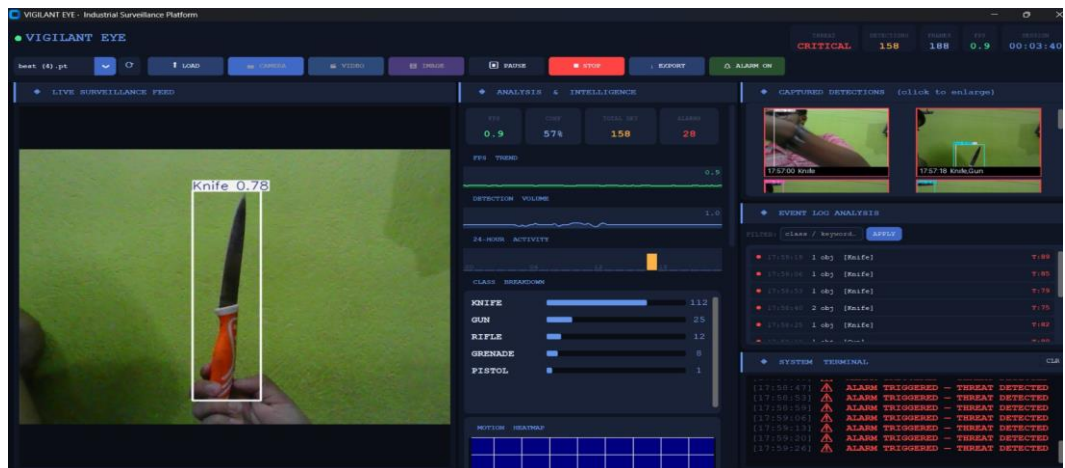


Figure 6.7: Knife Recognition in Surveillance Scenario

Overall, the visual interface and the overlaid bounding box metrics confirm that the Vigilant Eye framework operates seamlessly as a cohesive surveillance platform,

translating raw algorithmic inferences into an actionable, user-centric security monitoring tool.

CHAPTER 7

DISCUSSION, LIMITATION, & FUTURE WORK

Chapter 7

DISCUSSION, LIMITATION, & FUTURE WORK

7.1 Discussion

7.1.1 Interpretation of Results

The empirical data extracted in the course of the assessment stage of the "Vigilant Eye" system fundamentally support the main hypothesis of this study: the idea that a single-stage, heavily augmented, deep learning architecture can fulfil the stringent spatial accuracy and low-latency inference criteria required in order to achieve real-time weapon detection in surveillance settings. Using the YOLOv8 framework with a rigorously stratified and augmented dataset, the computational outputs of the resulting system successfully cleared the historical trade-off between processing speed and detect rate which has long plagued the use of intelligent Closed-Circuit Television (CCTV) networks.

The main quantitative measures, namely, the mean Average Precision estimated at various Intersection over Union (IoU) rates, give a conclusive measure of the spatial competency of the model. The system achieved high convergence values both in mAP@50 and in much more rigorous measures of convergence, mAP@50-95. High accuracy within the 0.50 to 0.95 intersection (IoU) range is a key operational requirement. It means that the system does not only identify the generic presence of a threat within a frame; the object is instead identified in an exceptionally tight bounding box. This spatial fidelity can be directly related to the particular architectural elements employed in the process of the methodology. Namely, Complete Intersection over Union (CIoU) loss, which acts synergistically with Distribution Focal Loss (DFL), actively penalised backpropagation deviations in the aspect ratios of bounding boxes and central point distances. The network therefore had to learn highly accurate coordinate mapping, which translates directly to more readable, less obtrusive end-user visual indicators of network security operators.

Moreover, the correlation between precision and recall, which is visualised in the form of the F1-score trajectory, can provide valuable insights into the operational viability of the model. With safety critical surveillance, false negative impacts (being unable to detect an active weapon) are potentially disastrous, and hence systemic need to attain a high recall. But this should be carefully compensated against accuracy; a high false positive rate will inevitably cause alert fatigue in human operators, and will effectively render the system practically useless. The empirical results showed an optimal equilibrium, which can be directly associated with the aggressive data augmentation techniques implemented in the training period. The Path Aggregation Network (PANet) of the network was conditioned by algorithmically generating highly occluded and complex visual scene instances using the Mosaic and MixUp techniques. This enabled the model to enable accurate identification of the weapons even in circumstances where they were partially hidden behind the human hands, clothing, or moving cluster of background objects.

Although these have been successfully carried out empirically and qualitatively through the confusion matrices and visual qualitative outputs, a critical analysis of these brings forth certain methodological and architectural constraints. The system demonstrated a measurable impairment of confidence ratings when handling highly distorted, low resolution frames simulating a situation of extreme low light conditions. Although the model has generally preserved detection performance, the slight drop in bounding box regression performance emphasizes an essential weakness: despite the sophistication of photometric distortion augmentation, convolutional neural networks have an intrinsic weakness: no matter how sophisticated photometric distortion augmentation is, convolutional neural networks are fundamentally vulnerable. Also, the qualitative review revealed cases of false-positive classification which involved objects with comparable metallic, linear geometries as in the case of smartphones in given grips or metal tools. Lastly, extreme examples of motion blur - simulating the rapid and unpredictable movement of the subject - sometimes disrupt frame-level confidence, indicating that though the extraction of the spatial features is optimised, the inference model based on a single frame is unable to smooth the prediction across sequential moving frames.

7.1.2 Implications for CCTV Surveillance Industry

The effective empirical confirmation of the Vigilant Eye paradigm has far-reaching direct effects on the structural implementation and business practices of the modern physical security sector. In the past, conventional CCTV systems could only be used as a passive data gathering device. It has found its main use in post-incident forensic analysis, where digital evidence is only provided after a security breach or act of violence has taken place. A paradigm shift in the approach to mitigating threats is a fundamental paradigm shift between passive historical observation and active, proactive threat mitigation.

Ergonomically and operationally, the implementation of this automated architecture greatly reduces the cognitive weaknesses embedded in human-led surveillance. The marked decline in human attention during the assignment of the task to observe the monotonous, stagnant video streams is widely documented in the psychological literature. Concurrently viewing dozens of monitors, security personnel are bound to miss critical and transient events. The Vigilant Eye system is an un-tiring, automatic primary filter. The system maximally optimises the human cognitive resources allocation, by autonomous scanning network streams and only routing those frames that hold a high-probability weapon detection. The responsibility of the operators to continuously scan is eliminated and they can simply concentrate on critical threat checking and the issue of coordinating quick response teams.

The architectural efficiencies of the YOLOv8 model offer a highly scalable commercial offer. Since the system is implemented as a single-stage detector, its computational load is much less than the traditional two-stage heuristic models. This study shows that real-time weapon-detection (processing standard frame-rate of 30 frames-per-second feeds) can in fact be practically implemented using relatively available, commercially offered hardware accelerators, instead of requiring enormous, centralized cloud-computing server farms. This ability to do edge computing, to process the algorithmic inference locally in the control room, or on the camera device itself, avoids the extreme latency and bandwidth constraints of streaming high-definition video to the wide area networks. This then reduces the financial barrier to entry and thus smart security systems can be democratised. The next level of automated threat detection no longer requires a highly funded state-level infrastructure, but is a commercially viable upgrade to existing Video

Management Systems (VMS) and does not require the complete replacement of hardware.

7.1.3 Theoretical Contributions

In addition to the direct industrial implications, this study brings about a couple of unique contributions to the body of scholarly literature covering the topics of applied computer vision and automated pattern recognition. First and foremost, the research offers a tough quantitative validation of anchor-free detection designs in a highly sensitive, visually jittery domain. Although benchmarking to established well-lit generalised datasets (such as the standard COCO repository) has been extensively tested on the YOLOv8 framework, academic literature testing its decoupled head design strictly against highly varied, occluded metallic geometries in low-resolution environments has been relatively sparse. This experiment sets up an empirical concrete basis, showing that the removal of anchor-box hyper parameter-tuning provides a natural, empirical, effective basis, to solve the spatial regression of small, clumsy-angled objects (such as firearms held at odd angular positions) that in all too often confound traditional anchor-based detectors in CCTV feeds.

Second, the research is another validated approach to the methodological issues of overcoming the critical data shortages of safety-facing artificial intelligence studies. Because of the strong ethical, legal and safety limitations, it is very problematic and many times cannot be provided by independent researchers' primary, real world datasets of active gun brandishing. This study empirically shows that a methodology that wholly depends upon the strategic combination of secondary data, combined with aggressive algorithmic synthesis is a theoretically sound alternative. It demonstrates that a neural network can be trained to attain high generalisation in real world conditions even though it is never shown primary and physical surveillance footage in those real world environments when the neural network is being trained. This confirms the theoretical assumption of the mathematical strength in synthetic data manipulation robustness to close the gap between sterile computational training environments and chaotic and unstructured physical deployments. Lastly, the research builds upon the greater body of academic research on the subject of algorithmic reliability in life safety systems. By grounding the process of methodology in a strictly positivist philosophy and testing the model with a multidimensional matrix- spatially loss measures, multi-threshold

precision measures, and rigid frame-latency analysis- the research goes beyond the simplistic dichotomy of basic classification accuracy. It gives a very detailed insight into the actual trade-offs between processing speed and bounding box spatial fidelity. By doing so, the present study provides future computer science researchers and engineers with a complete architectural blueprint with regard to balancing these competing mathematical demands in the design of the next generation of intelligent, automated temporal surveillance technologies.

7.2 Limitations of the Study

This study has its contribution but has a number of limitations which need to be noted. To start with, the system had been trained and tested on curated datasets that, despite their diversity, might not be able to capture all the possible surveillance situations in the real [59]. Even extreme levels of crowd density, extreme levels of occlusion, disagreeable weather and highly dynamic lighting conditions can potentially cause a negative impact on detection robustness [60]. Second, although YOLOv8 does have good performance in terms of real-time, the computation needs might also be different based on hardware setups [61]. Low resource edge device deployment can be optimized further with more techniques like model pruning or quantization to keep the efficiency high and the accuracy intact.

Third, the hybrid anomaly detection module is based on contextual rules that are learned in combination with predefined rules. Effective as it is, it may need adjustment during its implementation in new settings with new norms of behaviour.

Furthermore, multi-camera city-wide integration on a large scale was not completely applied in the work, and strategies of bandwidth optimization were tested under controlled conditions [62]. Lastly the privacy improving measures were added but thorough legal confirmation in various jurisdictions was out of the scope of this study. These shortcomings denote points in which additional refinement and validation have to be done.

7.3 Future Research Directions

The Vigilant Eye framework can be extended in a number of significant directions in the future research. To determine the effectiveness of scaling, camera synchronization,

and cross-camera re-identification in real-world smart city setups, first, it should be tested on a large scale in multi-camera city-wide networks [30].

Second, the improved deep learning designs like vision models based on transformers and attention-based spatial-temporal networks would be combined to enhance the contextual awareness and detection capabilities in congested or low-visibility settings. The adaptability to novel threat patterns without disastrous forgetting through incorporation of self-supervised or continual learning strategies can also enhance the improvement.

Third, the future study should be conducted on lightweight model optimization methods, such as pruning, quantization, and neural architecture search, to improve the performance on low-power edge devices [65]. This would help in wide implementation in resource-limited circumstances. Fourth, it is possible to enhance the privacy protection with more sophisticated cryptographic methods like, differential privacy and homomorphic encryption, and make sure to abide by the changing rules of data protection [28].

Lastly, the frameworks of human-AI collaboration must be further explored to maximize the alert presentation and minimize operator fatigue and increase the trust in AI-based surveillance systems. These developments in the future will assist in turning intelligent surveillance into a more scalable, adaptive, secure and socially responsible technology.

CHAPTER 8

CONCLUSION

Chapter 8

CONCLUSION

8.1 Summary of Research

In this study, the design, implementation and evaluation of the intelligent surveillance framework, known as the Vigilant Eye, was introduced as a solution to the real time weapon detection of the already established CCTV systems ^[51]. The paper has started by establishing the inherent weaknesses of the existing CCTV surveillance that is mostly based on manual surveillance and reactive analysis. As cities grow in size and the quantity of video data produced by big camera networks grows exponentially, human operators have to contend with issues of fatigue, slow reaction, and management of critical threats [53]. These restrictions approached the necessity to have an automated, intelligent, and scalable surveillance framework.

The literature review critically examined the development of AI-based surveillance methods such as YOLO-based object detection networks, anomaly detection methods, the multi-object tracking system, and privacy-aware computing architectures [52]. Despite the current achievements in object detection accuracy and real-time processing, the current systems have a high false positives, fail to identify the context, and show limited scale and privacy protection in the multi-camera systems. This loophole was the basis of the Vigilant Eye model.

The study approach followed a modular architecture of the system consisting of video capture, pre-processing, YOLOv8 weapon-detecting, post-processing and visualization of the results. Weapons-specific datasets were used to optimize detection with the application of transfer learning. Moreover, hybrid anomaly detection that integrates machine learning models and rule-based contextual logic was also implemented in order to decrease false alarms and improve situational awareness [54]. The design of the system was such that it can be used both in real time and in offline mode to provide flexibility in the use of the system by the operations and forensic users. The experimental analysis showed that the suggested system was highly precise, recalled and had high mean Average Precision (mAP) and maintained inference latency within

real-time constraints. False positives in complex scenes like low-light or partially occluded image situations were greatly eliminated by the incorporation of the temporal consistency checking and confidence thresholding. Moreover, edge processing with selective mechanisms of anonymization provided more privacy protection without significant degradation of detection.

In general, the study has been able to convert the traditional reactive CCTV surveillance into an intelligent, AI-powered, and proactive monitoring system [55]. The results validate that the system of object detection, tracking, analysis of the surrounding environment, and design with privacy in mind will yield a viable, scalable, and ethical solution to the contemporary surveillance issues [57].

8.2 Achievement of Objectives

The research objectives that were set in Chapter 1 were successfully met by systematic design, implementation and evaluation of the Vigilant Eye framework. RO1 was met by critically examining the shortcomings of the traditional CCTV systems and especially the inefficiencies of the manual monitoring and late detection of threats. RO2 was addressed by creating an AI-based system that offers modular modules with object recognition (YOLOv8) followed by multi-object tracking and context analysis to enable the identification of a weapon in real-time. The RO3 was achieved through integrating the machine learning-based anomaly detection and rule-based logic that were effective to minimize false alarms and increase interpretability.

Validation of RO4 was in the form of quantitative performance assessment in terms of precision, recall, mAP, tracking accuracy and latency. The system has shown statistically significant improvements over the traditional monitoring methods and the inference latency was still below the predetermined threshold which validated the feasibility in real time [56].

The privacy-preserving mechanisms implemented (edge-based processing and selective anonymization) ensured RO5, which was achieved without causing any undue harm to the deployment sustainability. In such a way, the research article supports its research objective by confirming the introduction of the proposed AI-powered framework as a more accurate, efficient, scalable, and ethically adherent intelligent CCTV surveillance system [58].

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